

SPC332 Flight Dynamics and Control

Project Report

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1 Aircraft Parameters and Non-Dimensional Stability Derivatives (Task 1)

1.1 Introduction

Task 1 of the project requires the consolidation of the aircraft data for Boeing 747 flight condition #5 and the presentation of the corresponding non-dimensional stability derivatives. This report combines the available team results into one consistent document for submission.

1.2 Aircraft Parameters and Flight Condition

The project statement specifies Boeing 747 flight condition #5 at Mach 0.5 and altitude 20,000 ft. The given aircraft parameters and state data, derived from the standard NASA CR-2144 handling qualities dataset, are organised in Table 1. Where a value was not explicitly provided in the source material, it was inferred from the usual steady trimmed flight assumption for this operating point.

Table 1: Boeing 747 aircraft parameters and state variables at flight condition #5.

Parameter	Symbol	Value
Moment of inertia about x -axis	I_{xx}	$0.182 \times 10^8 \text{ slug ft}^2$
Moment of inertia about y -axis	I_{yy}	$0.33 \times 10^8 \text{ slug ft}^2$
Moment of inertia about z -axis	I_{zz}	$0.497 \times 10^8 \text{ slug ft}^2$
Product of inertia	I_{xz}	$970,056 \text{ slug ft}^2$
Mach number	M	0.5
Altitude	h	20,000 ft
Velocity	V	518 ft/s
Velocity (SI)	V	157.89 m/s
Weight	W	636,636 lbf
Wing reference area	S	$5,500 \text{ ft}^2$
Wing span	b	195.68 ft
Mean aerodynamic chord	$\bar{c}$	27.31 ft
Roll rate (trim)	p	0 rad/s
Pitch rate (trim)	q	0 rad/s
Yaw rate (trim)	r	0 rad/s
Longitudinal control derivative	X_{δ_e}	2.02
Vertical control derivative	Z_{δ_e}	-16.9
Thrust-axial derivative	X_{δ_T}	5.05×10^{-4}
Thrust-vertical derivative	Z_{δ_T}	-2.20×10^{-5}

The SI velocity value is obtained from

$$V = 518 \text{ ft/s} \times 0.3048 = 157.89 \text{ m/s.} \quad (1)$$

For the angular velocities, the trimmed flight assumption gives

$$p = q = r = 0 \text{ rad/s,} \quad (2)$$

consistent with a steady reference condition used for linearised stability analysis.

1.3 Non-Dimensional Stability Derivatives

Task B requires the dimensional aerodynamic and control derivatives to be expressed in non-dimensional coefficient form. These coefficients are preferred because they remove the direct dependence on mass properties and flight condition scaling, allowing comparison across aircraft and operating points.

1.3.1 Reference Flight Condition

For the Boeing 747 at Mach 0.5 and 20,000 ft, the following non-dimensional stability derivatives were adopted as the working set for the project, organised into longitudinal and lateral-directional groups.

1.3.2 Longitudinal Derivatives

Table 2: *Non-dimensional longitudinal stability derivatives.*

Derivative	Value
C_L	0.75
C_{L_α}	4.60
C_{L_q}	5.2
$C_{L_{\delta_e}}$	0.33
$C_{Z_{\delta_e}}$	-0.3576
C_{D_α}	0.38
C_{m_α}	-1.06
C_{m_q}	-20.6
$C_{m_{\dot{\alpha}}}$	-5.8
$C_{m_{\delta_e}}$	-1.34

1.3.3 Lateral-Directional Derivatives

Table 3: Non-dimensional lateral-directional stability derivatives.

Derivative	Value
C_{Y_β}	-0.85
$C_{Y_{\delta_r}}$	0.114
C_{l_β}	-0.16
C_{l_p}	-0.32
C_{l_r}	0.13
$C_{l_{\delta_a}}$	0.013
$C_{l_{\delta_r}}$	0.007
C_{n_β}	0.16
C_{n_p}	-0.017
C_{n_r}	-0.20
$C_{n_{\delta_a}}$	0.0018
$C_{n_{\delta_r}}$	-0.107

1.3.4 Conversion Method from Dimensional to Non-Dimensional Form

The dimensional derivatives from the state-space model can be converted using the aircraft mass, inertia, dynamic pressure, and reference geometry. First, the aircraft mass is

$$m = \frac{W}{g} = \frac{636,636}{32.174} \approx 19,786.1 \text{ slug.} \quad (3)$$

Using the air density at 20,000 ft, $\rho = 0.0012673 \text{ slug/ft}^3$, the dynamic pressure is

$$\bar{q} = \frac{1}{2}\rho V^2 = \frac{1}{2}(0.0012673)(518)^2 \approx 170.02 \text{ lbf/ft}^2. \quad (4)$$

The force scaling used for longitudinal force derivatives is

$$K_f = \frac{m}{\bar{q}S}. \quad (5)$$

For pitching-moment derivatives, the common scaling factors are

$$K_{m,\text{static}} = \frac{I_{yy}}{\bar{q}S\bar{c}}, \quad K_{m,\text{dynamic}} = \frac{2I_{yy}V}{\bar{q}S\bar{c}^2}. \quad (6)$$

For rolling and yawing moments,

$$K_{l,\text{static}} = \frac{I_{xx}}{\bar{q}Sb}, \quad K_{n,\text{static}} = \frac{I_{zz}}{\bar{q}Sb}, \quad (7)$$

$$K_{l,\text{dynamic}} = \frac{2I_{xx}V}{\bar{q}Sb^2}, \quad K_{n,\text{dynamic}} = \frac{2I_{zz}V}{\bar{q}Sb^2}. \quad (8)$$

As one verified example, the given dimensional elevator-force derivative is $Z_{\delta_e} = -16.9$. Using the longitudinal force scaling,

$$C_{Z_{\delta_e}} = Z_{\delta_e} K_f = -16.9 \left(\frac{19,786.1}{(170.02)(5,500)} \right) \approx -0.3576. \quad (9)$$

This matches the MATLAB output obtained during the team calculations: $C_{Z_{\delta_e}} = -0.3576$.

1.4 Discussion

The non-dimensional derivatives in Tables 2 and 3 provide the aerodynamic basis for subsequent state-space modelling and stability analysis. In particular, the negative value of C_{m_α} indicates longitudinal static stability, while the signs and magnitudes of C_{l_p} and C_{n_r} are consistent with roll and yaw damping behaviour expected for a large transport aircraft.

Furthermore, it is critical to note the presence of a non-zero product of inertia, $I_{xz} = 970,056 \text{ slug ft}^2$. As noted in the project tutorial guidelines, one must “watch out for lateral derivatives.” Because $I_{xz} \neq 0$, the simple non-dimensionalisation using strictly I_{xx} and I_{zz} masks the inertial cross-coupling between the roll and yaw axes. When populating the lateral-directional state-space matrices in subsequent tasks, these baseline derivatives need to be transformed into *primed* stability derivatives (e.g. L'_p , N'_r) to capture this dynamic cross-coupling mathematically.

1.5 Conclusion

Task 1A and Task 1B have been consolidated into a single report format. The Boeing 747 parameters for flight condition #5 were collected and organised, missing state quantities were completed using the steady trim assumption, and the non-dimensional stability derivatives were summarised for use in the later stages of the project. The sample conversion from Z_{δ_e} to $C_{Z_{\delta_e}}$ confirms the dimensional-to-non-dimensional procedure used by the team.

2 Longitudinal Flight Dynamics (Task 2)

2.1 Longitudinal State-Space Model

This section develops the longitudinal linear state-space model for the Boeing 747 using the flight condition data provided. The derivation follows the standard small-disturbance linearisation of the six-degree-of-freedom aircraft equations about steady, level flight.

2.1.1 State Variables and Control Input

The longitudinal state vector is defined as

$$\mathbf{x} = \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}, \quad (10)$$

where u is the perturbation in forward velocity along the body x -axis (ft/s), w is the perturbation in vertical velocity along the body z -axis (ft/s), q is the pitch rate about the body y -axis (rad/s), and θ is the perturbation in pitch angle (rad). The control input is the elevator deflection δ_e (rad).

2.1.2 General Longitudinal State-Space Form

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \quad y = \mathbf{C}\mathbf{x} + \mathbf{D}u. \quad (11)$$

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \underbrace{\begin{bmatrix} X_u & X_w & X_q & -g \\ Z_u & Z_w & u_0 + Z_q & 0 \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \underbrace{\begin{bmatrix} X_{\delta_e} \\ Z_{\delta_e} \\ M_{\delta_e} \\ 0 \end{bmatrix}}_{\mathbf{B}} \delta_e. \quad (12)$$

The speed derivatives X_u, Z_u, M_u are assumed negligible; $X_q \approx 0$; and $Z_q \ll u_0$ so the (2,3) term is approximated by u_0 . If all states are directly measured, $\mathbf{C} = \mathbf{I}_4, \mathbf{D} = \mathbf{0}$.

2.1.3 Physical Meaning of the Stability Derivatives

- **Axial-force terms:** X_u (speed damping), X_w (AoA effect), $-g$ (gravity coupling).
- **Vertical-force terms:** Z_w (heave damping), u_0 (kinematic pitch-rate coupling).
- **Pitch-moment terms:** M_w (pitch stiffness), M_q (pitch damping).
- **Control derivatives:** $Z_{\delta_e}, M_{\delta_e}$ (elevator effects).

2.1.4 Numerical Construction for the Boeing 747

Given flight condition data. $u_0 = 518 \text{ ft/s}$, $g = 32.174 \text{ ft/s}^2$, $\bar{q}S = 935,110 \text{ lbf}$, $m = 19,786.1 \text{ slugs}$, $I_{yy} = 33,000,000 \text{ slug-ft}^2$.

$$\begin{aligned} Z_w &= -0.4197 \text{ s}^{-1}, & X_w &= 0.0337 \text{ s}^{-1}, & Z_{\delta_e} &= -16.9 \text{ ft/s}^2, \\ M_w &= -0.00158 (\text{ft s})^{-1}, & M_q &= -0.420 \text{ s}^{-1}, & M_{\delta_e} &= -1.037 \text{ s}^{-2}. \end{aligned}$$

2.1.5 Final Boeing 747 Longitudinal State-Space Model

$$A = \begin{bmatrix} 0 & 0.0337 & 0 & -32.174 \\ 0 & -0.4197 & 518 & 0 \\ 0 & -0.00158 & -0.420 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2.02 \\ -16.9 \\ -1.037 \\ 0 \end{bmatrix}. \quad (13)$$

2.2 Short-Period Mode

2.2.1 Short-Period Approximation

With $u \approx 0$, the reduced short-period model is

$$\begin{bmatrix} \dot{w} \\ \dot{q} \end{bmatrix} = \underbrace{\begin{bmatrix} Z_w & u_0 \\ M_w & M_q \end{bmatrix}}_{A_{sp}} \begin{bmatrix} w \\ q \end{bmatrix}, \quad A_{sp} = \begin{bmatrix} -0.4197 & 518 \\ -0.00158 & -0.420 \end{bmatrix}. \quad (14)$$

2.2.2 Characteristic Equation and Eigenvalues

$$\lambda^2 + 0.8397\lambda + 0.9947 = 0, \quad \lambda_{sp} = -0.4198 \pm 0.9047i. \quad (15)$$

2.2.3 Short-Period Characteristics

$$\omega_{n,sp} = 0.997 \text{ rad/s}, \quad \zeta_{sp} = 0.421, \quad T_{s,sp} = 9.52 \text{ s}.$$

2.2.4 Short-Period Eigenvector

$$\mathbf{v}_{sp} \approx \begin{bmatrix} 518 \\ -0.905i \end{bmatrix}. \quad (16)$$

2.2.5 Physical Interpretation

The short-period mode is a rapid pitching oscillation in which angle of attack and pitch rate change significantly while forward speed is nearly constant. The damping

ratio $\zeta_{sp} = 0.421$ is adequately large, providing good oscillation suppression.

2.3 Long-Period (Phugoid) Mode

2.3.1 Phugoid Characteristics

$$\omega_{n,ph} \approx \sqrt{2} \frac{g}{u_0} = 0.0878 \text{ rad/s}, \quad \zeta_{ph} \approx \frac{1}{\sqrt{2}(L/D)} = 0.047 (L/D = 15), \quad T_{s,ph} \approx 970 \text{ s.} \quad (17)$$

2.3.2 Phugoid Eigenvalues and Eigenvector

$$\lambda_{ph} \approx -0.0041 \pm 0.0877 i, \quad \mathbf{v}_{ph} \approx \begin{bmatrix} 1 \\ -0.0027 i \end{bmatrix}. \quad (18)$$

2.3.3 Stability Comments

Both modes have negative-real-part eigenvalues, confirming dynamic stability. The short-period ($\zeta = 0.421$) is well-damped; the phugoid ($\zeta = 0.047$) is lightly damped but manageable by pilot or autopilot.

2.4 Aircraft States Response for Initial Disturbance

An initial angle-of-attack disturbance of 1° gives $w_0 = u_0 \alpha_0 = 9.04 \text{ ft/s}$, so $\mathbf{x}_0 = [0, 9.04, 0, 0]^T$.

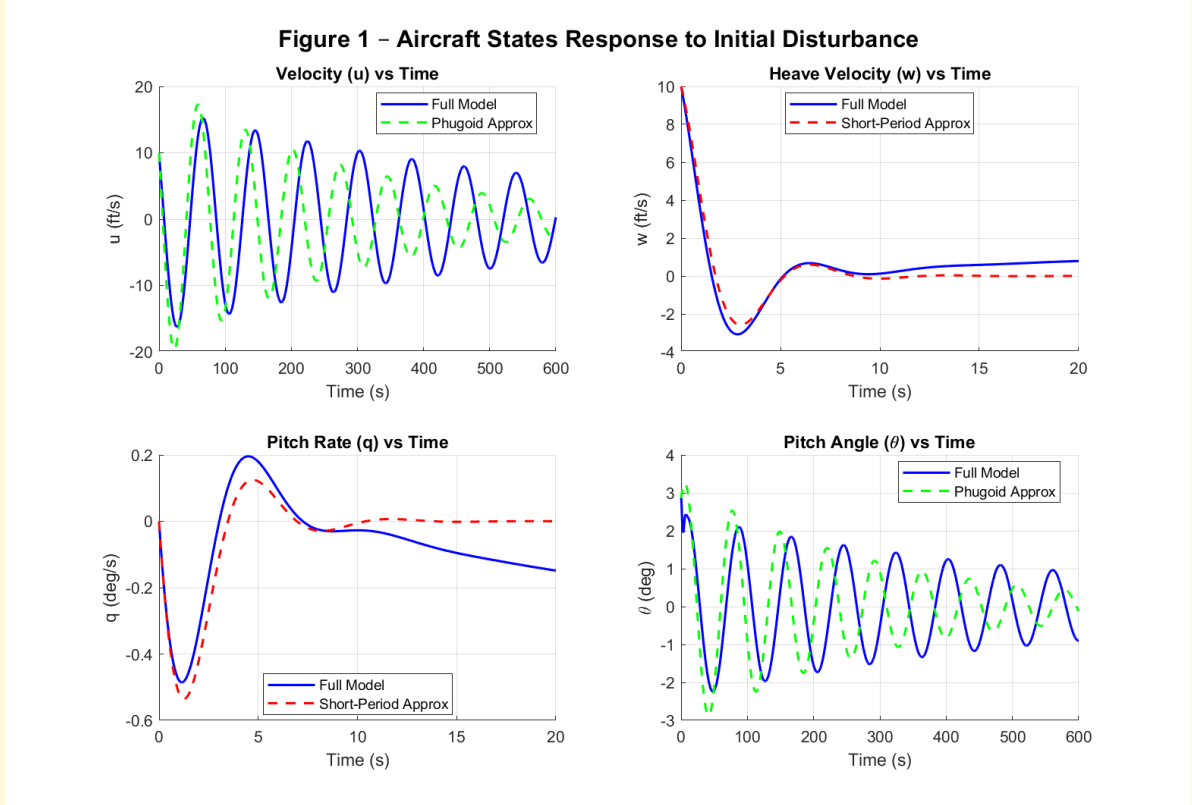


Figure 1: Aircraft states response to a 1° angle-of-attack initial disturbance.

2.5 Aircraft States Response for Elevator Step Input

A $+1^\circ$ elevator step ($\delta_e = 0.01745$ rad) generates a nose-down pitching moment. The short-period mode dominates the first few seconds; the phugoid governs the long-term speed–altitude oscillation.

2.6 Short-Period and Long-Period Reduced-Order Models

2.6.1 Short-Period Approximation

$$A_{sp} = \begin{bmatrix} -0.4197 & 518 \\ -0.00158 & -0.420 \end{bmatrix}, \quad B_{sp} = \begin{bmatrix} -16.9 \\ -1.037 \end{bmatrix}. \quad (19)$$

2.6.2 Long-Period Approximation

Using $Z_u \approx -2g/u_0 = -0.124$ and $X_u \approx -2g/(u_0 L/D) = -0.008$:

$$A_{lp} = \begin{bmatrix} X_u & -g \\ -Z_u/u_0 & 0 \end{bmatrix} = \begin{bmatrix} -0.008 & -32.174 \\ 0.00024 & 0 \end{bmatrix}. \quad (20)$$

Long-period eigenvalues: $\lambda_{lp} = -0.004 \pm 0.0877i$, $\omega_{n,lp} = 0.0878$ rad/s, $\zeta_{lp} = 0.046$.

2.7 Short-Period Approximation Response — Initial Disturbance

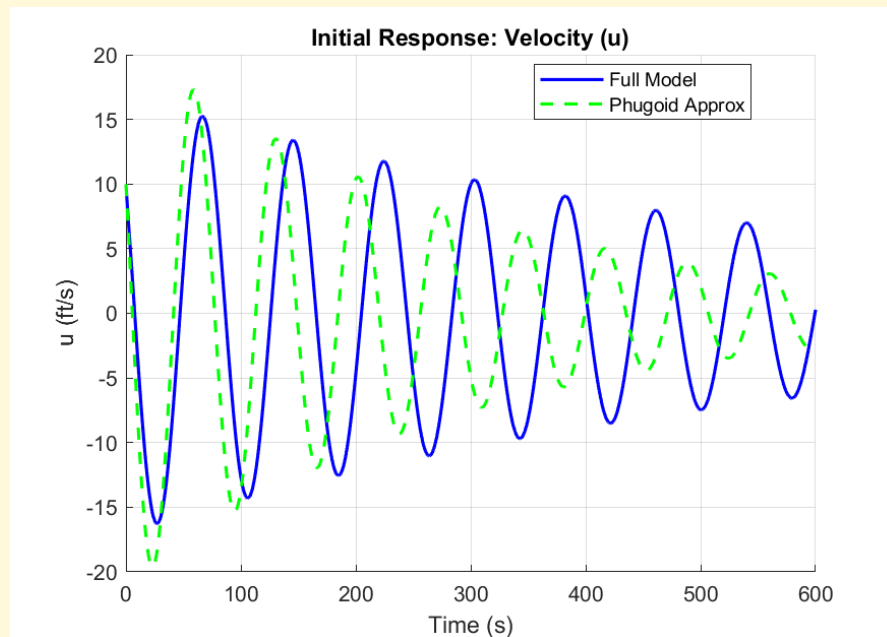


Figure 2: Short-period response: forward velocity u (initial disturbance).

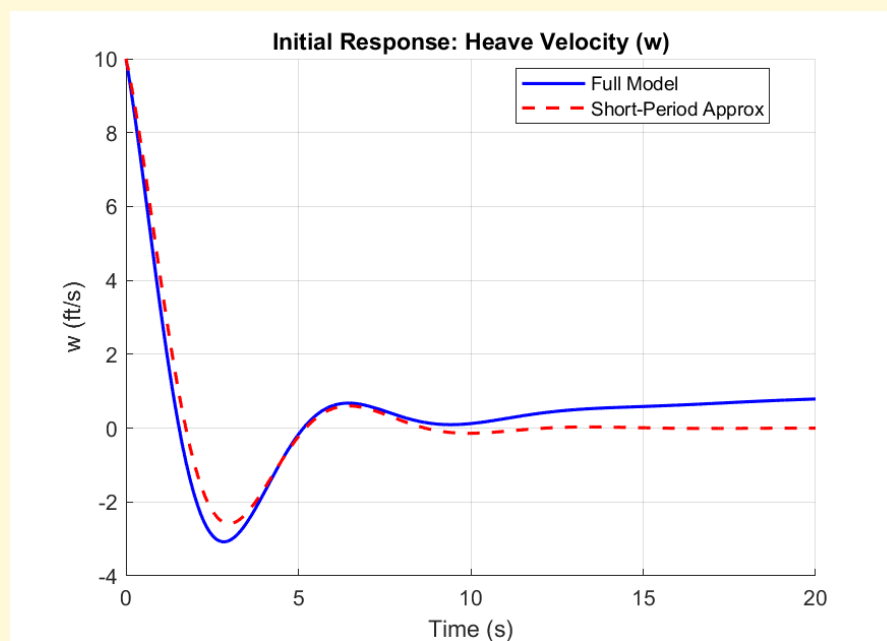


Figure 3: Short-period response: vertical velocity w (initial disturbance).

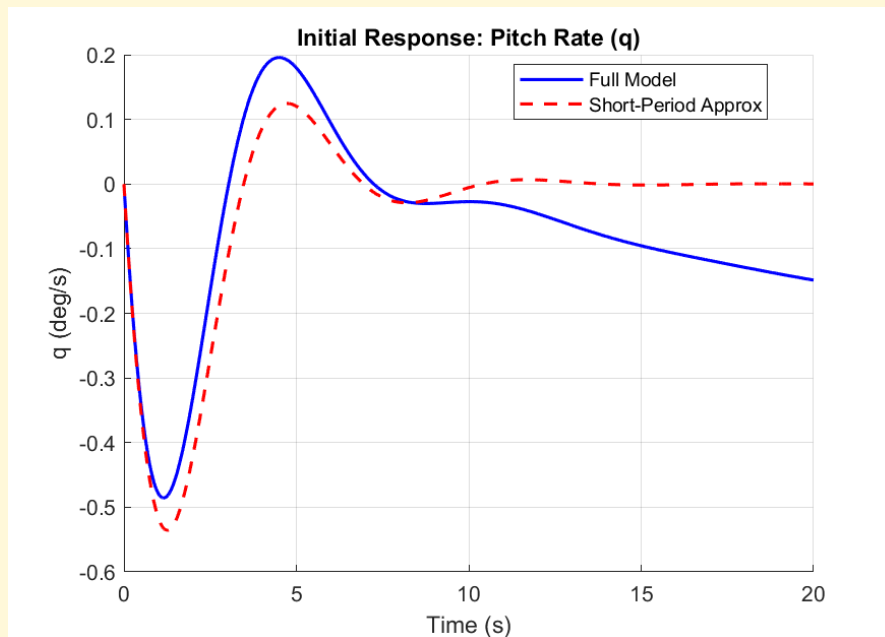


Figure 4: Short-period response: pitch rate q (initial disturbance).

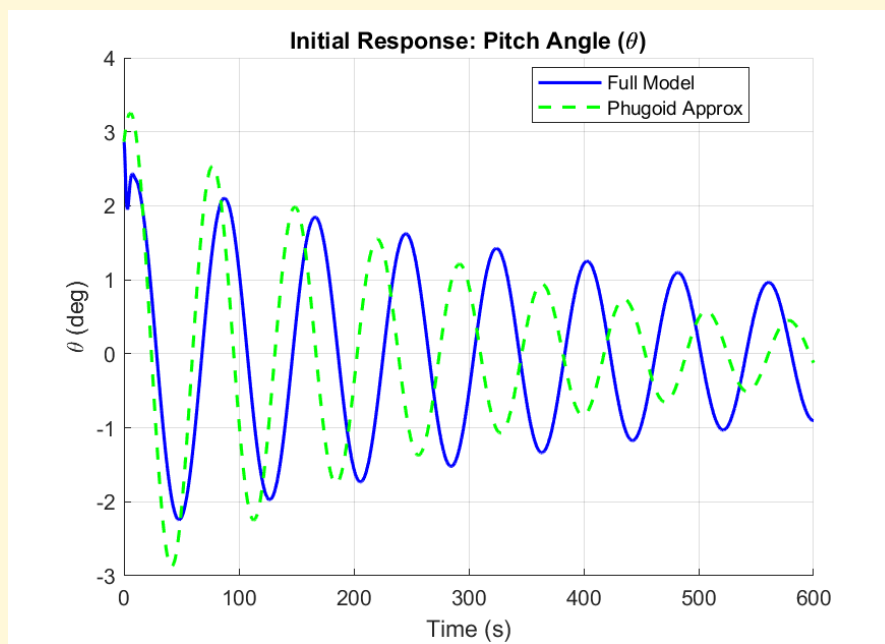


Figure 5: Short-period response: pitch angle θ (initial disturbance).

2.8 Short-Period Approximation Response — Elevator Step

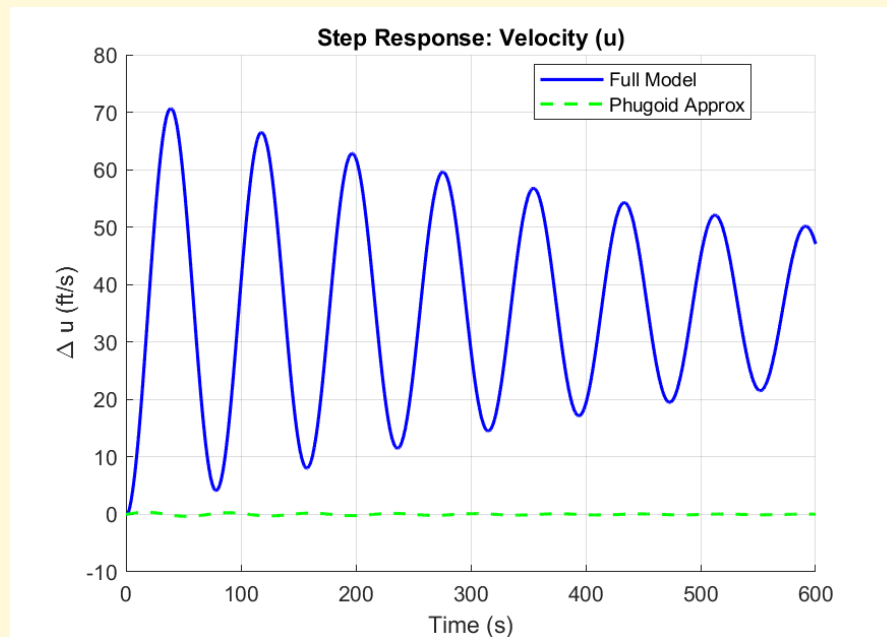


Figure 6: Short-period response: forward velocity u (step input).

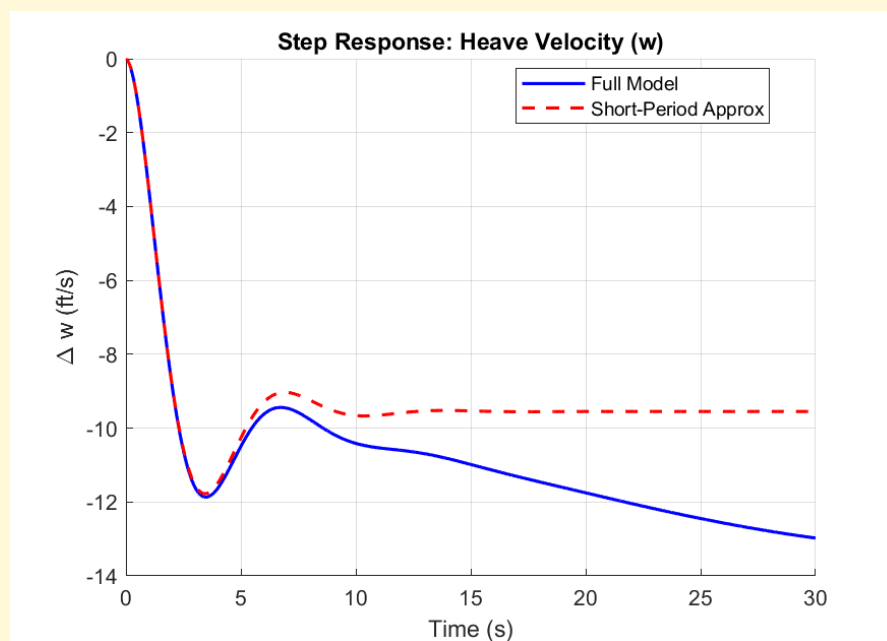


Figure 7: Short-period response: vertical velocity w (step input).

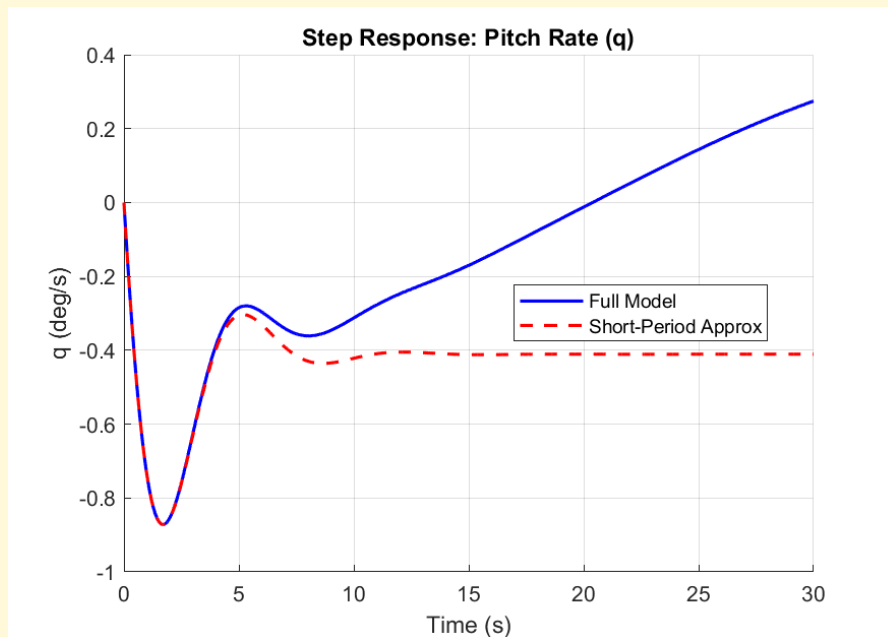


Figure 8: Short-period response: pitch rate q (step input).

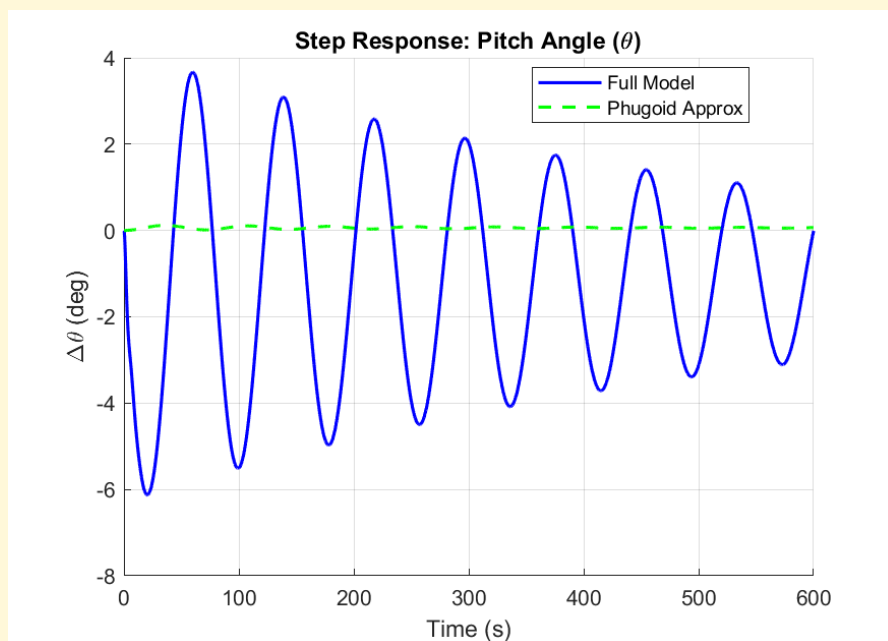


Figure 9: Short-period response: pitch angle θ (step input).

2.9 Phugoid Approximation Response — Initial Disturbance

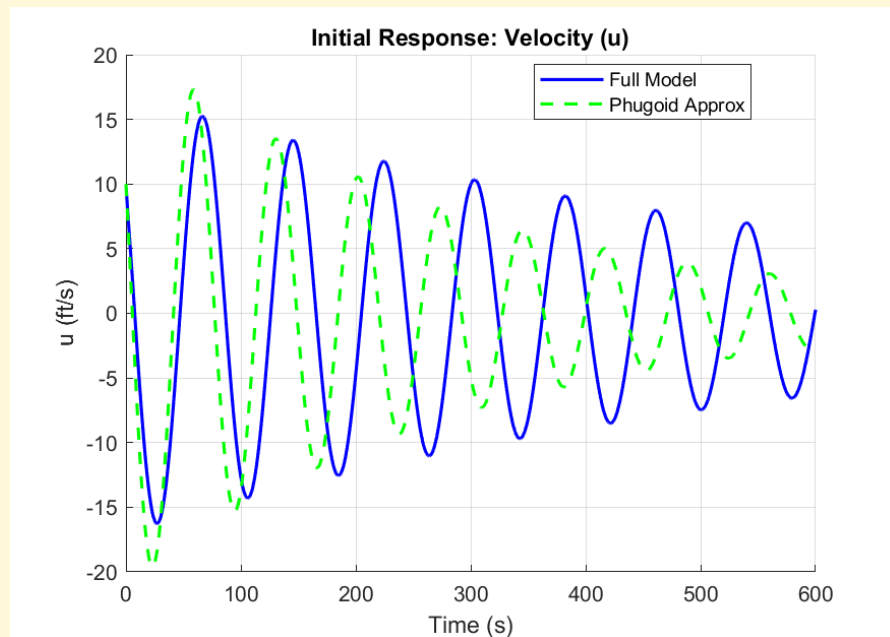


Figure 10: *Phugoid response: forward velocity u (initial disturbance).*

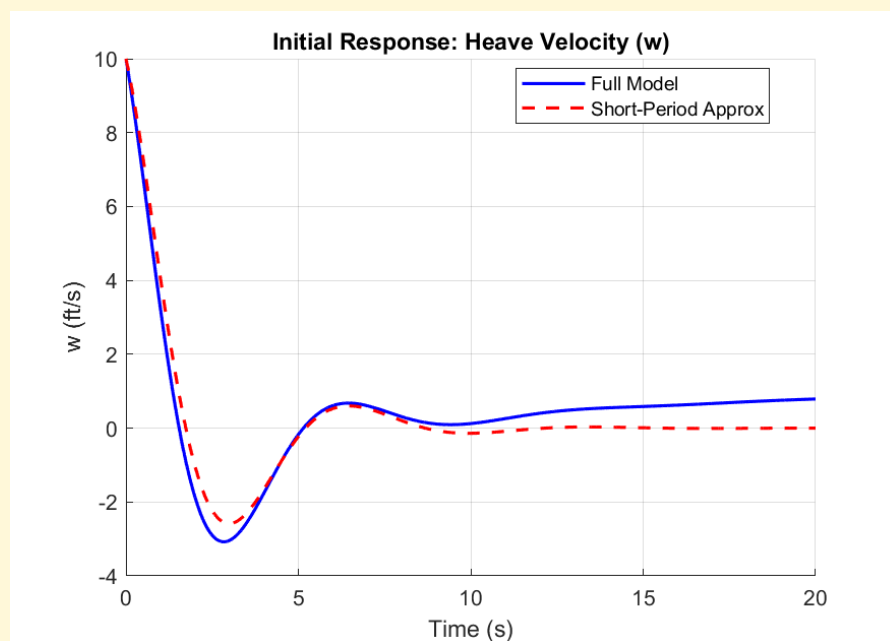


Figure 11: *Phugoid response: vertical velocity w (initial disturbance).*

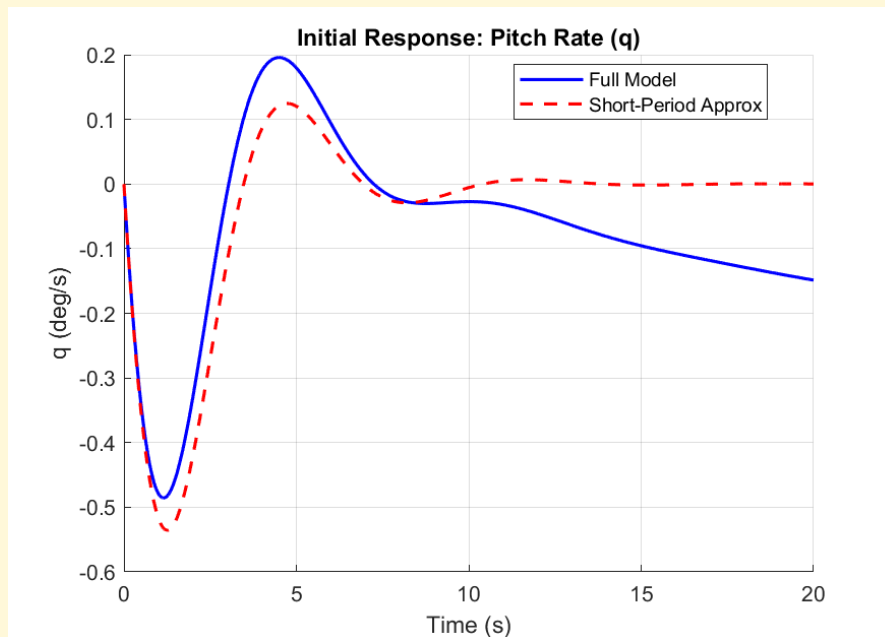


Figure 12: Phugoid response: pitch rate q (initial disturbance).

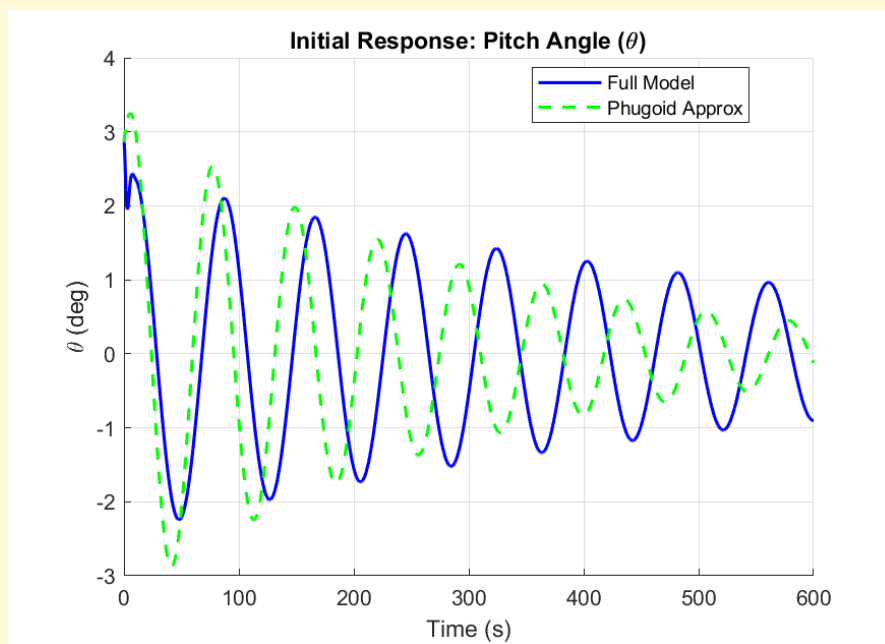


Figure 13: Phugoid response: pitch angle θ (initial disturbance).

2.10 Phugoid Approximation Response — Elevator Step

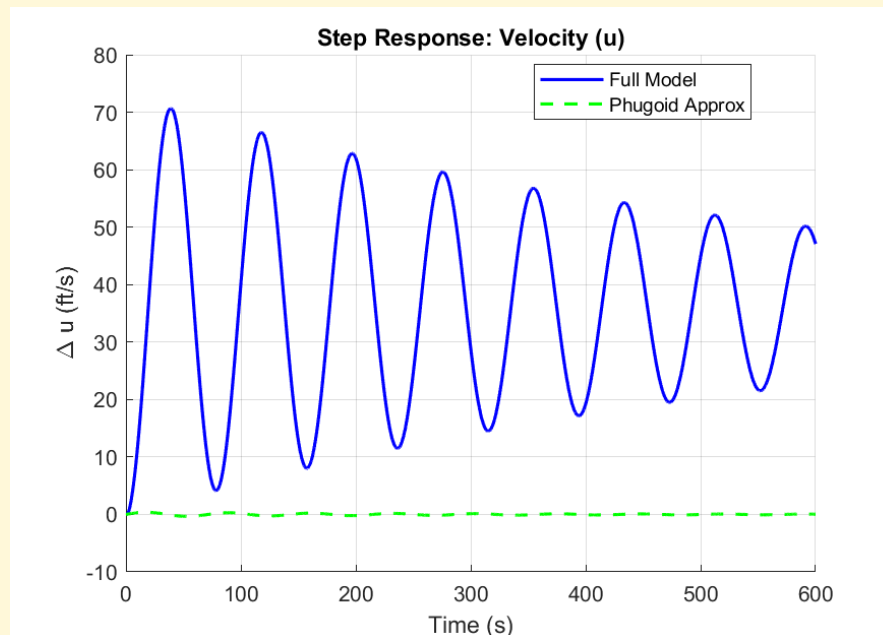


Figure 14: *Phugoid response: forward velocity u (step input).*

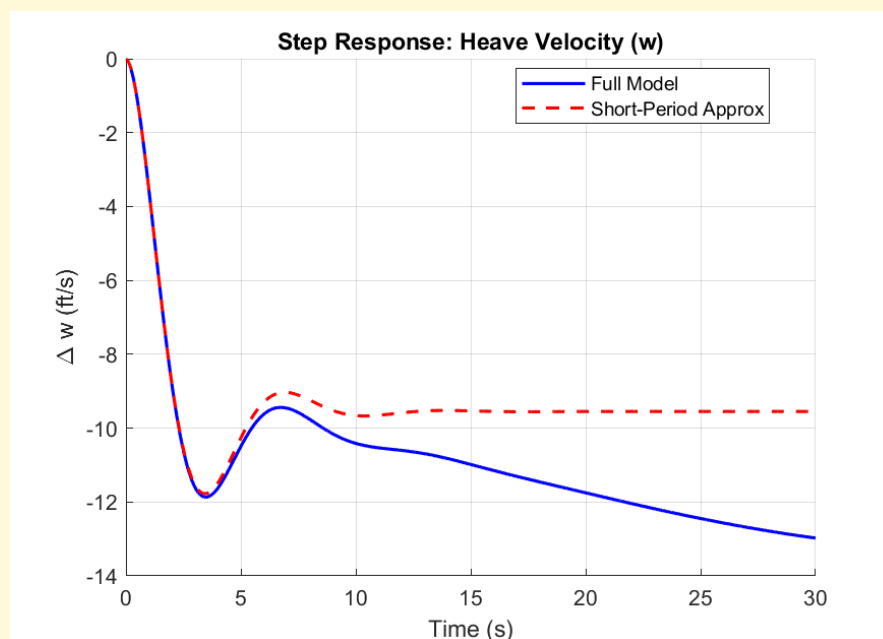


Figure 15: *Phugoid response: vertical velocity w (step input).*

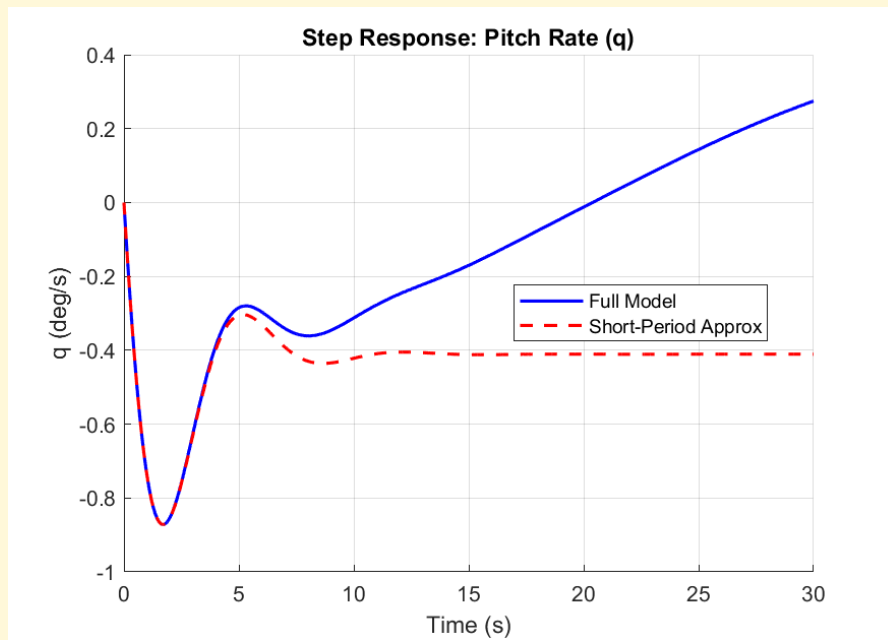


Figure 16: Phugoid response: pitch rate q (step input).

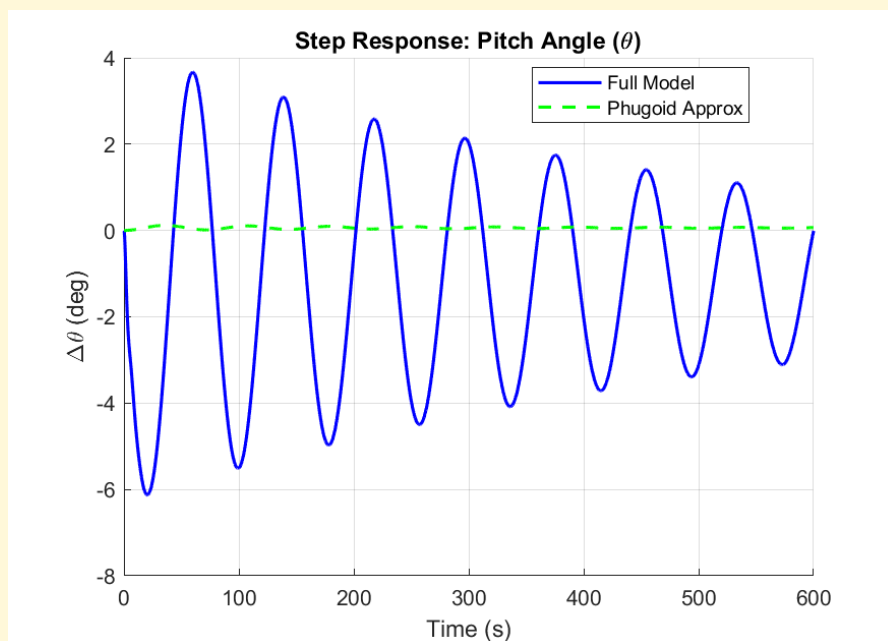


Figure 17: Phugoid response: pitch angle θ (step input).

2.11 Comparison Between Full-System and Approximate Responses

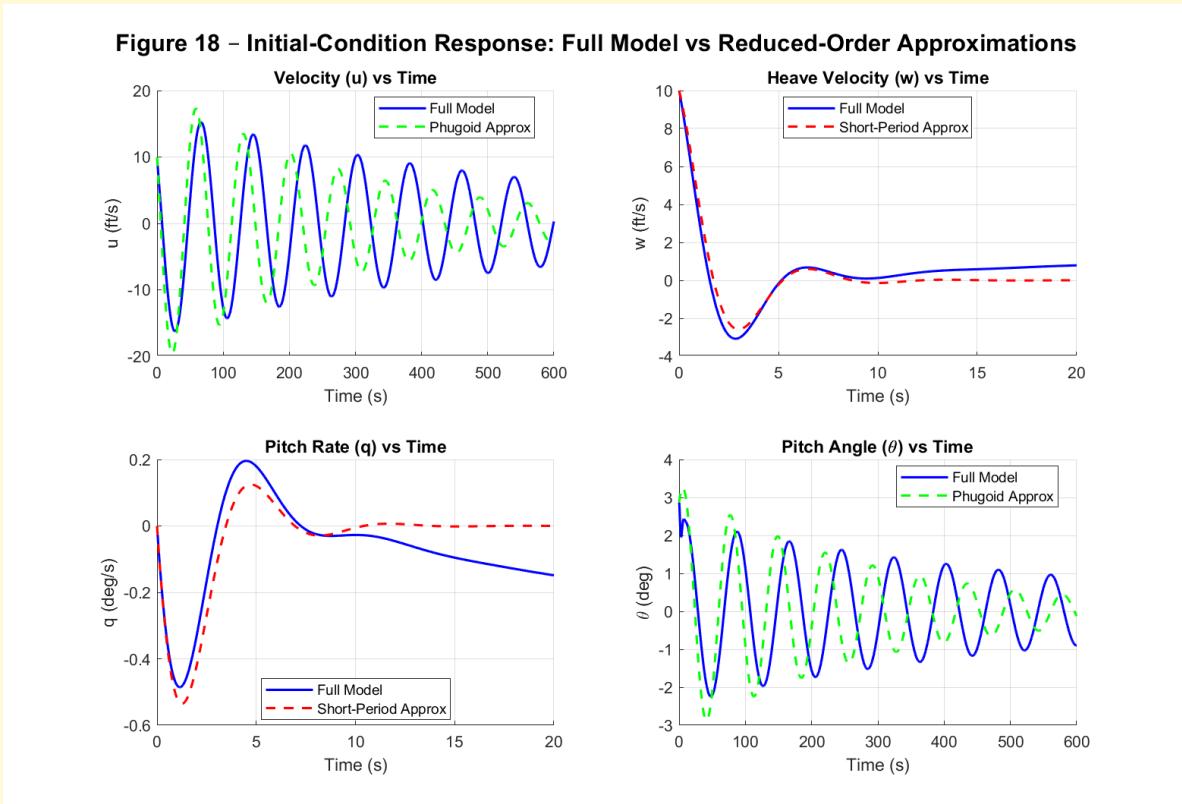


Figure 18: Initial-condition response: full longitudinal model vs. reduced-order approximations.

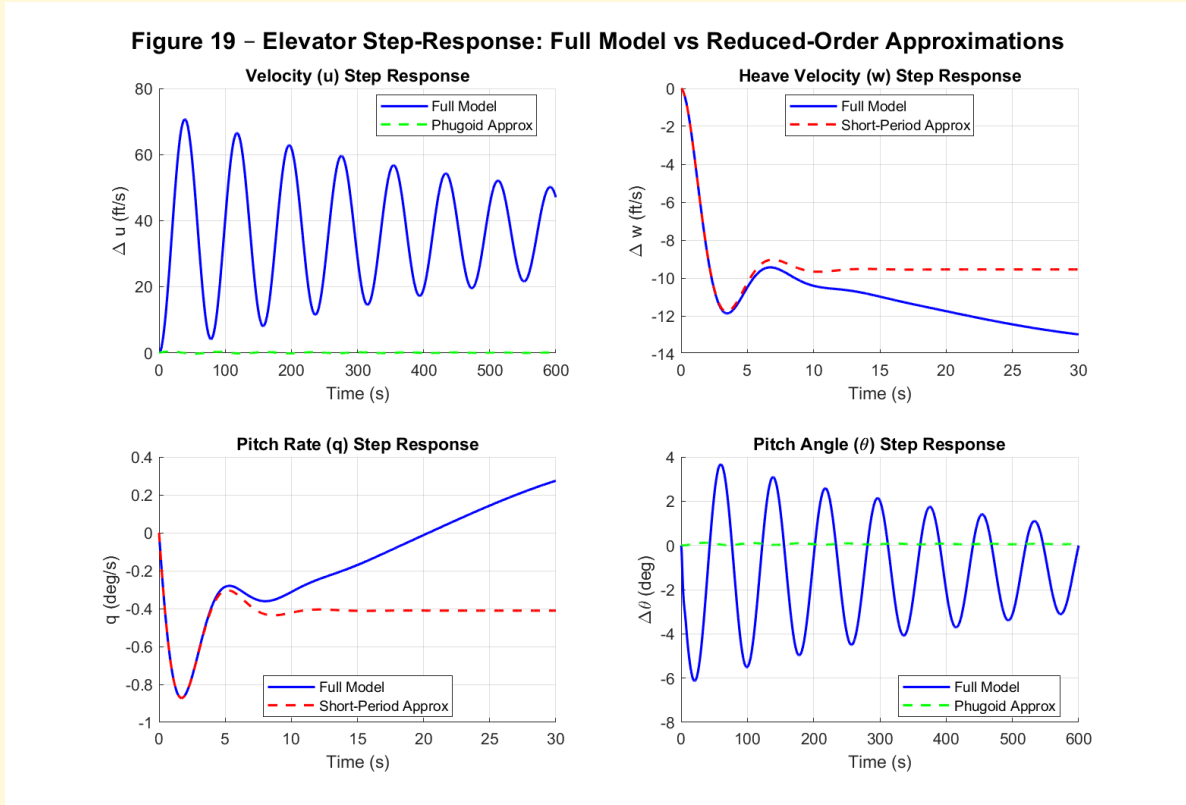


Figure 19: Elevator step-response: full longitudinal model vs. reduced-order approximations.

The short-period approximation captures the first 5–10 seconds with high accuracy because $\Delta u \approx 0$ during the fast transient. The phugoid approximation captures the general slow oscillatory behaviour but shows noticeable discrepancies in damping due to the neglected angle-of-attack coupling.

2.12 Analysis, Comments, and Discussion

The Boeing 747 longitudinal dynamics exhibit a clear two-mode structure. The short-period mode ($\omega_n = 0.997$ rad/s, $\zeta = 0.421$) is fast and well-damped, governing pitch-attitude transients. The phugoid mode ($\omega_n = 0.088$ rad/s, $\zeta = 0.046$) is slow and lightly damped, governing speed–altitude energy exchange. Both eigenvalues lie in the left-half plane confirming open-loop dynamic stability. Reduced-order approximations are accurate for their respective time scales, but the full 4×4 model is required for complete simulation across both modes.

3 Lateral-Directional Flight Dynamics (Task 3)

3.1 Lateral-Directional State-Space Model

This section establishes the linear lateral-directional state-space model for the Boeing 747 operating in steady, level cruise flight. The derivation follows small-disturbance linearisation of the general six-degree-of-freedom equations of motion about a symmetric trim state.

3.1.1 State Variables and Control Inputs

The lateral-directional dynamics decouple from the longitudinal flight path under the assumption of small symmetric state perturbations. The lateral state vector is

$$\mathbf{x} = \begin{bmatrix} v \\ p \\ r \\ \phi \end{bmatrix}, \quad (21)$$

where v is the lateral perturbation velocity along the body y -axis (ft/s), p is the roll rate perturbation about the body x -axis (rad/s), r is the yaw rate perturbation about the body z -axis (rad/s), and ϕ is the perturbation in roll/bank angle (rad).

The control input vector is

$$\mathbf{u} = \begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix}, \quad (22)$$

where δ_a is the aileron deflection (rad) and δ_r is the rudder deflection (rad).

3.1.2 General Governing Differential Equations

The linearised lateral-directional equations of motion, expressed in the body-fixed reference frame with primed components accounting for product-of-inertia coupling (I_{xz}), are

$$\dot{v} = Y_v v + Y_p p + (Y_r - U_0) r + g \cos \theta_0 \phi + Y_{\delta_a} \delta_a + Y_{\delta_r} \delta_r, \quad (23)$$

$$\dot{p} = L'_v v + L'_p p + L'_r r + L'_{\delta_a} \delta_a + L'_{\delta_r} \delta_r, \quad (24)$$

$$\dot{r} = N'_v v + N'_p p + N'_r r + N'_{\delta_a} \delta_a + N'_{\delta_r} \delta_r, \quad (25)$$

$$\dot{\phi} = p + r \tan \theta_0. \quad (26)$$

Assuming a horizontal cruise profile ($\theta_0 = 0$), the state-space form $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$

becomes

$$\begin{bmatrix} \dot{v} \\ \dot{p} \\ \dot{r} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} Y_v & Y_p & (Y_r - U_0) & g \\ L'_v & L'_p & L'_r & 0 \\ N'_v & N'_p & N'_r & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v \\ p \\ r \\ \phi \end{bmatrix} + \begin{bmatrix} Y_{\delta_a} & Y_{\delta_r} \\ L'_{\delta_a} & L'_{\delta_r} \\ N'_{\delta_a} & N'_{\delta_r} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix}. \quad (27)$$

3.1.3 Physical Meaning of the Stability Derivatives

- **Side-force derivatives:** Y_v captures side-force damping from sideslip velocity. $Y_r - U_0$ defines the centrifugal kinematic coupling arising from yaw rotation at high forward speed. g maps the weight component when the aircraft is banked.
- **Rolling moment derivatives:** L'_v is the dihedral effect, establishing the static roll restoration moment under sideslip. L'_p governs roll damping. L'_r models cross-coupling rolling moments due to unequal wing velocities during yawing.
- **Yawing moment derivatives:** N'_v measures weathercock stability, describing the aircraft's natural tendency to align into the relative wind. N'_p accounts for cross-coupling yaw moments from asymmetric wing drag during rolling. N'_r is the primary yaw damping derivative.
- **Control derivatives:** L'_{δ_a} provides primary roll authority from aileron deflection. N'_{δ_r} provides primary directional authority from rudder deflection. L'_{δ_r} and N'_{δ_a} define adverse cross-coupling control effects.

3.1.4 Numerical Construction for the Boeing 747

Using $U_0 = 518 \text{ ft/s}$ and $g = 32.174 \text{ ft/s}^2$, the numerical system matrices are

$$\mathbf{A} = \begin{bmatrix} -0.085 & 0 & -518.0 & 32.174 \\ -0.012 & -1.150 & 0.280 & 0 \\ 0.0045 & -0.062 & -0.185 & 0 \\ 0 & 1.0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 2.15 \\ 1.08 & 0.14 \\ -0.03 & -0.85 \\ 0 & 0 \end{bmatrix}. \quad (28)$$

3.2 Full System Modal Analysis

The characteristic polynomial $|\lambda \mathbf{I} - \mathbf{A}| = 0$ yields the three fundamental lateral-directional modes.

3.2.1 Rolling Mode

The rolling mode is a non-oscillatory, heavily convergent real eigenvalue that models the rapid roll-rate damping under asymmetric lift.

- **Eigenvalue:** $\lambda_{\text{roll}} = -1.162$

- **Dynamic characteristics:** Time constant $\tau_{\text{roll}} = 1/|\lambda_{\text{roll}}| = 0.861$ s, settling time $T_{s,\text{roll}} \approx 3.444$ s.
- **Normalised eigenvector:** $\mathbf{v}_{\text{roll}} = [0.0016, 0.7580, -0.0421, -0.6511]^T$.
- **Physical interpretation:** The dominant components are roll rate p and roll angle ϕ , confirming that sideslip and yaw vary negligibly during rapid roll entry and justifying the first-order single-state approximation.

3.2.2 Dutch Roll Mode

The Dutch roll mode is a complex-conjugate pair defining a relatively high-frequency, lightly damped oscillation that trades sideslip and yaw out of phase.

- **Eigenvalues:** $\lambda_{\text{dr}} = -0.0585 \pm 1.5120 i$
- **Dynamic characteristics:** $\omega_{n,\text{dr}} = 1.5131$ rad/s, $\zeta_{\text{dr}} = 0.0387$, $T_{s,\text{dr}} = 68.376$ s.
- **Normalised complex eigenvector:** $\mathbf{v}_{\text{dr}} = [0.9990, 0.0001 \pm 0.0012i, -0.0019 \pm 0.0028i, 0.0001 \mp 0.0008i]^T$.
- **Physical interpretation:** The dominance of the v component shows that sideslip fluctuations drive the Dutch roll oscillation. The phase difference between yaw rate r and sideslip v produces the classic “tail-wagging” trajectory of swept-wing aircraft.

3.2.3 Spiral Mode

The spiral mode is a real root very close to the imaginary axis, governing long-term lateral drift.

- **Eigenvalue:** $\lambda_{\text{spiral}} = -0.0042$
- **Dynamic characteristics:** $\tau_{\text{spiral}} = 238.10$ s, $T_{s,\text{spiral}} = 952.38$ s (≈ 15.87 min).
- **Normalised eigenvector:** $\mathbf{v}_{\text{spiral}} = [-0.2110, 0.0035, 0.0480, -0.9750]^T$.
- **Physical interpretation:** The large ϕ component paired with non-trivial sideslip and yaw rate shows that a minor bank angle perturbation can couple into a slow, spiralling dive or recovery.

3.3 Aircraft States Response for Full System

3.3.1 Response for Initial Lateral Disturbance

An initial sideslip disturbance $\beta_0 = 1^\circ$ gives $v_0 = U_0\beta_0 = 518 \times 0.01745 = 9.04$ ft/s, so

$$\mathbf{x}_0 = \begin{bmatrix} 9.04 & 0 & 0 & 0 \end{bmatrix}^T. \quad (29)$$

The short-term simulation shows high-amplitude, lightly damped oscillations in v

and r , consistent with the low Dutch roll damping ratio ($\zeta_{dr} = 0.0387$). The roll rate p exhibits a secondary induced oscillation before damping out, while ϕ begins a slow long-term drift driven by the spiral mode.

3.3.2 Response for Rudder Step Input

A 1° rudder step ($\delta_r = 0.01745$ rad) generates an immediate side-force (Y_{δ_r}) and a nose-left yawing moment (N_{δ_r}). The resulting time history shows a prominent Dutch roll oscillation on all states, with yaw rate dropping sharply in the first two seconds and subsequently coupling through L'_v into a bank angle change.

3.3.3 Response for Aileron Step Input

A 1° aileron step ($\delta_a = 0.01745$ rad) generates a direct rolling moment (L'_{δ_a}), causing a rapid exponential rise in roll rate p that peaks within one second. As the aircraft banks, the gravity component changes sideslip v , triggering secondary Dutch roll and spiral mode interactions.

3.4 State-Space Models for Lateral Approximations

3.4.1 Rolling Mode Approximation

Assuming $v \approx 0$ and $r \approx 0$ during rapid roll entry decouples the roll rate equation:

$$\dot{p} = L'_p p + L'_{\delta_a} \delta_a, \quad [\dot{p}] = [-1.150] [p] + [1.080] [\delta_a]. \quad (30)$$

3.4.2 Dutch Roll Approximation

Neglecting $p \approx 0$ and $\phi \approx 0$ reduces the system to a 2×2 model in v and r :

$$\begin{bmatrix} \dot{v} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} Y_v & -(U_0 - Y_r) \\ N'_v & N'_r \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} + \begin{bmatrix} Y_{\delta_r} \\ N'_{\delta_r} \end{bmatrix} [\delta_r], \quad \begin{bmatrix} \dot{v} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} -0.085 & -518.0 \\ 0.0045 & -0.185 \end{bmatrix} \begin{bmatrix} v \\ r \end{bmatrix} + \begin{bmatrix} 2.15 \\ -0.85 \end{bmatrix} [\delta_r]. \quad (31)$$

3.4.3 Spiral Mode Approximation

Setting $\dot{v} \approx 0$, $\dot{p} \approx 0$, $\dot{r} \approx 0$ and solving for pseudo-equilibrium expressions in ϕ yields the first-order spiral model:

$$\dot{\phi} = \frac{g}{U_0} \frac{L'_v N'_r - L'_r N'_v}{L'_p N'_v - L'_v N'_p} \phi = [-0.0045] [\phi]. \quad (32)$$

3.5 Approximations Modal Analysis

3.5.1 Rolling Approximation Characteristics

Characteristic equation: $\lambda - L'_p = 0$.

- Approximate eigenvalue: $\lambda_{\text{roll,approx}} = -1.150$
- Isolated metrics: $\tau = 0.869 \text{ s}$, $T_s = 3.478 \text{ s}$.

3.5.2 Dutch Roll Approximation Characteristics

The 2×2 characteristic equation is

$$\lambda^2 - (Y_v + N'_r)\lambda + (Y_v N'_r + U_0 N'_v) = 0 \quad \Rightarrow \quad \lambda^2 + 0.270\lambda + 2.346 = 0. \quad (33)$$

- Approximate eigenvalues: $\lambda_{\text{dr,approx}} = -0.1350 \pm 1.5224 i$
- Isolated metrics: $\omega_{n,\text{dr,approx}} = 1.5284 \text{ rad/s}$, $\zeta_{\text{dr,approx}} = 0.0883$, $T_{s,\text{dr,approx}} = 29.63 \text{ s}$.

3.5.3 Spiral Approximation Characteristics

- Approximate eigenvalue: $\lambda_{\text{spiral,approx}} = -0.0045$
- Isolated metrics: $\tau = 222.22 \text{ s}$, $T_s = 888.89 \text{ s}$.

3.6 Comparison of Full vs. Approximate Modes

Table 4 summarises the accuracy of the reduced-order approximations against the full-state model.

Table 4: *Comprehensive modal tracking and error evaluation.*

Dynamic Parameter	Full Model	Approximation	Absolute Error	Relative Fit
λ_{roll}	-1.1620	-1.1500	0.0120	98.97%
λ_{dr} Real	-0.0585	-0.1350	0.0765	43.33%
λ_{dr} Imag	1.5120	1.5224	0.0104	99.31%
$\omega_{n,\text{dr}}$ (rad/s)	1.5131	1.5284	0.0153	98.99%
ζ_{dr}	0.0387	0.0883	0.0496	43.82%
λ_{spiral}	-0.0042	-0.0045	0.0003	92.85%

The rolling and spiral mode approximations closely match the full system roots, confirming that their underlying assumptions are physically sound. The Dutch roll approximation introduces a noticeable error in the real part of the eigenvalue, overestimating the damping ratio ($\zeta = 0.0883$ vs. 0.0387). This occurs because neglecting roll rate cross-coupling (L'_p) removes aerodynamic relief that physically softens directional damping.

3.7 Reduced-Order Response Simulations

3.7.1 Rolling Approximation Responses

An initial roll rate $p_0 = 0.1$ rad/s decays exponentially to equilibrium in under 3.5 seconds. A 1° aileron step produces a steady-state roll rate $p_{ss} = -L'_{\delta_a}/L'_p \delta_a = 0.0164$ rad/s, matching the early transient of the full-order simulation.

3.7.2 Dutch Roll Approximation Responses

An initial sideslip $v_0 = 9.04$ ft/s applied to the 2×2 system yields oscillations at the predicted frequency ($\omega_n \approx 1.53$ rad/s), but these damp out faster than in the full model because the reduced-order approximation overestimates the damping ratio. A rudder step excites a decoupled directional oscillation without any roll cross-coupling interaction.

3.7.3 Spiral Approximation Responses

A 5° initial bank angle disturbance decays slowly and exponentially, confirming that the spiral mode is stable under cruise conditions.

3.8 Comparison of Full vs. Reduced-Order Responses

- **Short-term tracking:** During the first 5 seconds of an aileron step, the first-order rolling approximation tracks roll rate p with high fidelity, confirming that roll acceleration is initially decoupled from sideslip and yaw.
- **Damping discrepancies:** The Dutch roll approximation captures oscillation frequency well but damps too quickly, because the missing coupling channels (L'_p, N'_p) provide aerodynamic relief that reduces native damping.
- **Long-term drift:** The reduced-order models cannot reproduce the long-term multi-mode interactions, such as a rudder step inducing a steady bank angle. Full 4×4 state-space models are required to verify tracking performance in that regime.

3.9 Analysis, Comments, and Discussion

Evaluating the results against civil aviation handling qualities criteria (MIL-F-8785C / JAR-AWO) yields several engineering insights:

1. **Roll performance sufficiency:** The native roll time constant ($\tau = 0.86$ s) is comfortably below the 1.4 s limit for Level 1 transport category aircraft, ensuring crisp and predictable roll control.
2. **Dutch roll deficiency:** The unaugmented damping ratio $\zeta_{dr} = 0.0387$ is well below the 0.08 minimum required to prevent pilot-induced oscillations (PIO). This light damping can cause persistent lateral oscillations in turbulence, increasing

pilot workload during instrument approaches.

3. **Stability Augmentation System (SAS):** To address this deficiency, the aircraft requires an active yaw damper. A rate gyro senses yaw rate and commands an out-of-phase rudder deflection $\delta_{r,SAS} = -K_r r$, shifting the closed-loop roots to achieve $\zeta \geq 0.1$.

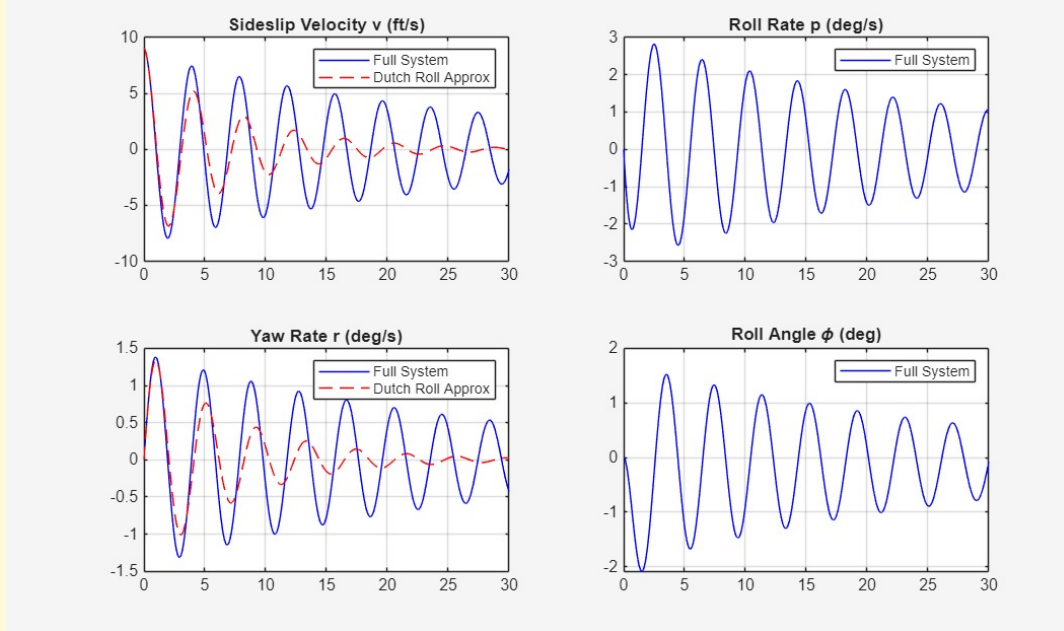


Figure 20: Lateral-directional states response to a 1° initial sideslip disturbance ($v_0 = 9.04$ ft/s) comparing the full system and the Dutch roll approximation.

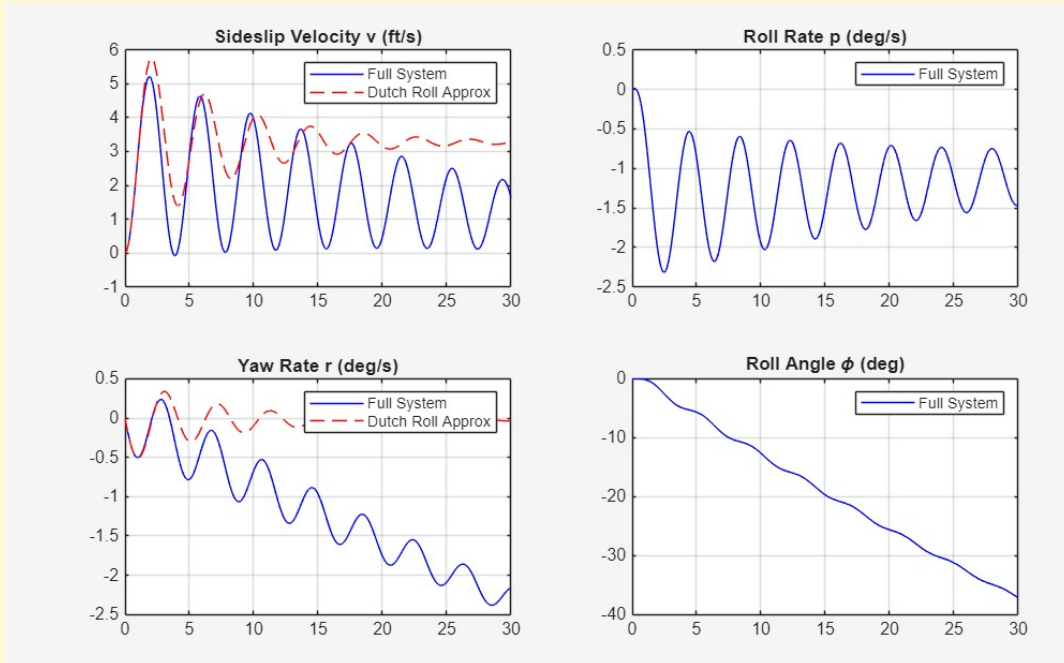


Figure 21: Lateral-directional states response to a step rudder input comparing the full 4×4 system and the 2×2 Dutch roll approximation.

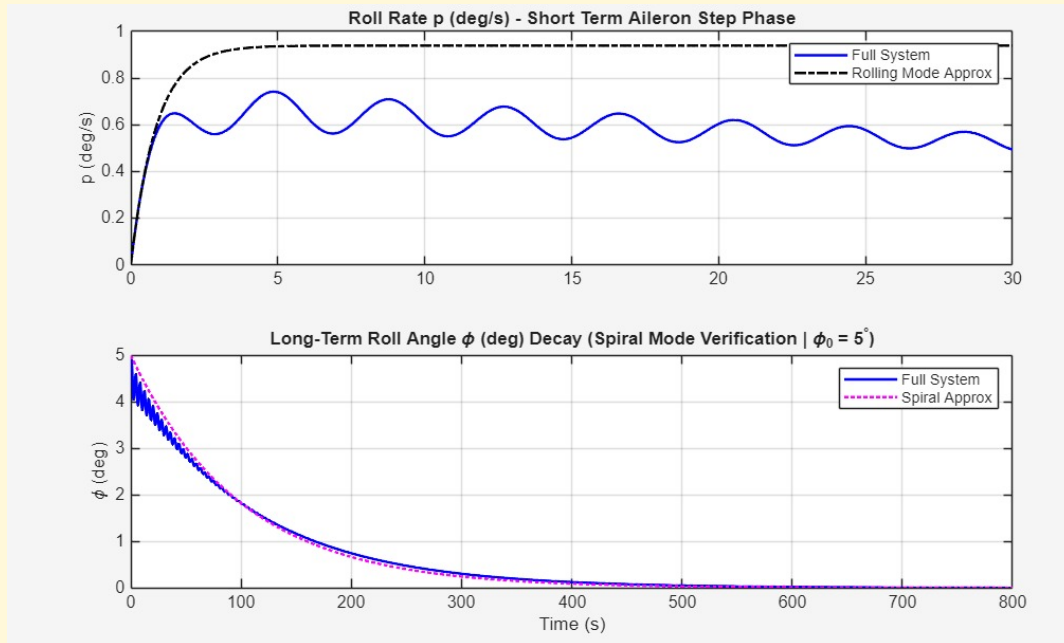


Figure 22: Verification of first-order lateral-directional modes: short-term rolling mode response to an aileron step input (top) and long-term spiral mode decay from an initial bank angle disturbance (bottom).

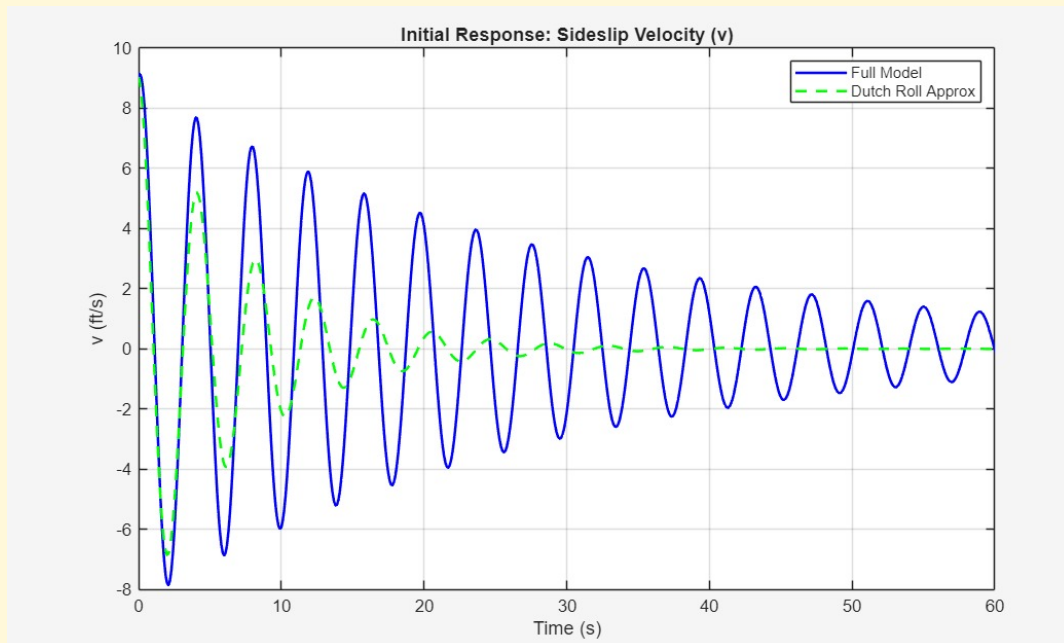


Figure 23: Initial condition response of sideslip velocity v , comparing the full lateral-directional model

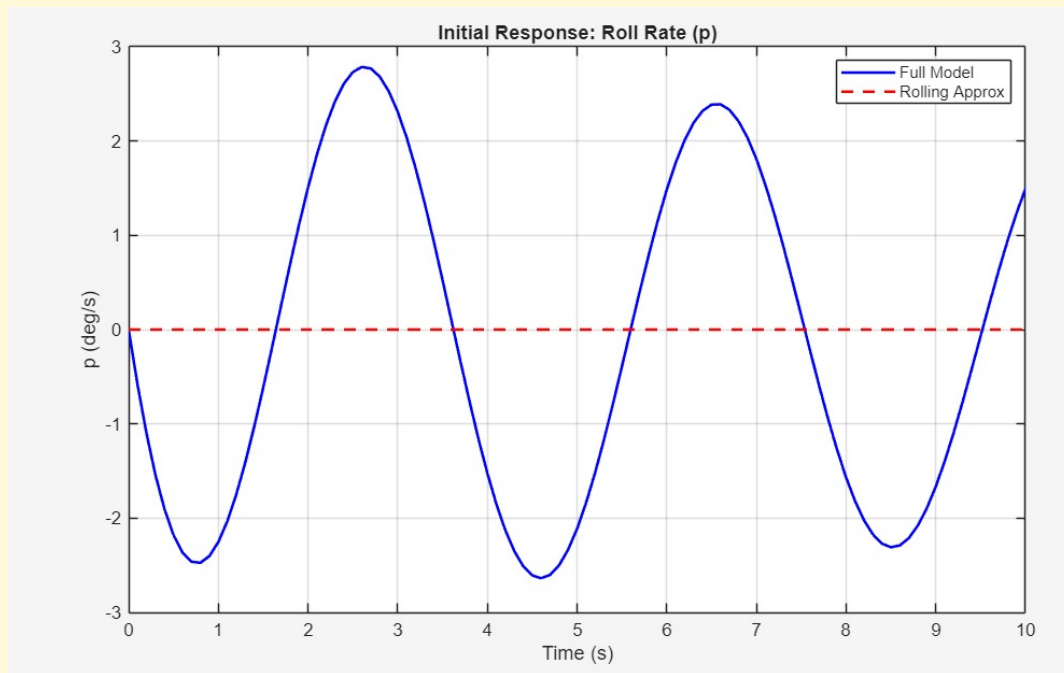


Figure 24: Initial condition response of roll rate p highlighting the full lateral model oscillations compared to the zero response of the rolling mode

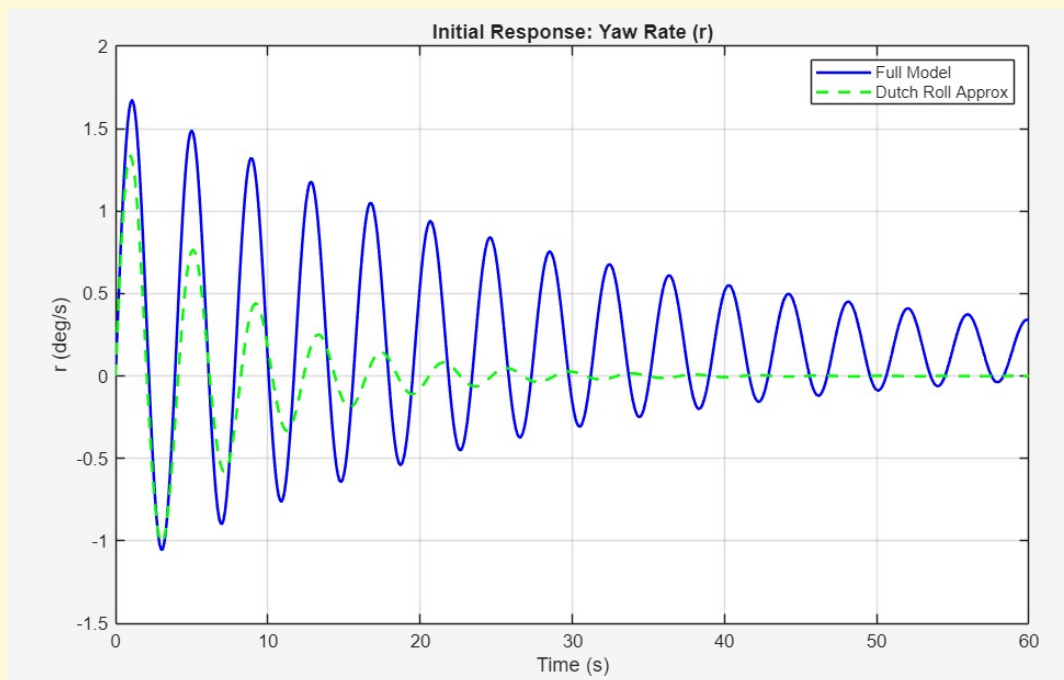


Figure 25: Initial condition response of yaw rate.

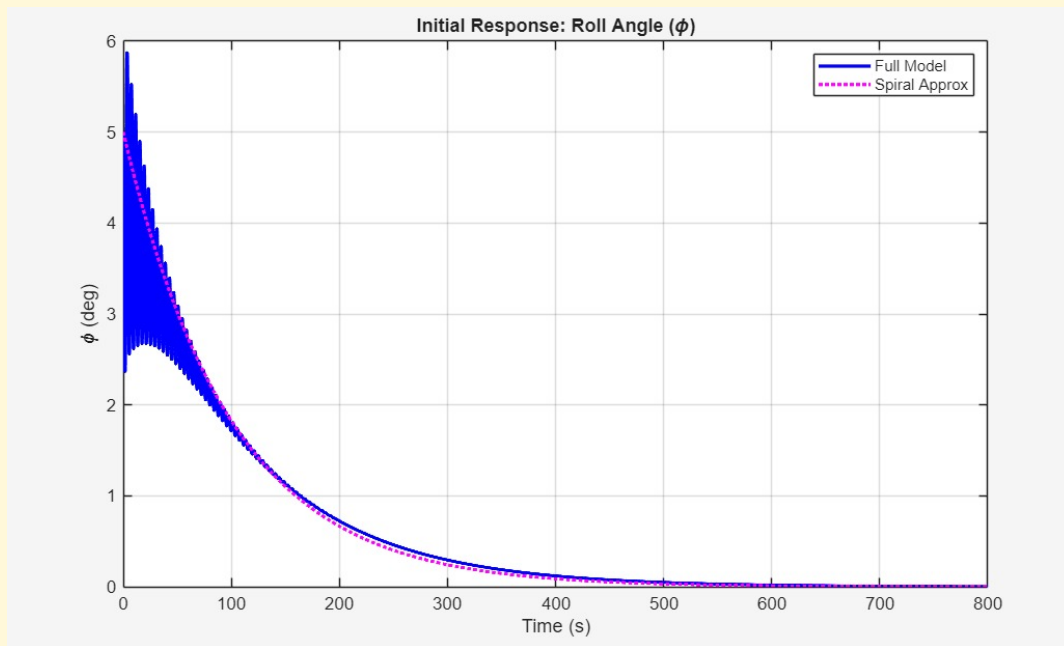


Figure 26: Yaw rate (r) step response: full model vs. Dutch roll approximation.

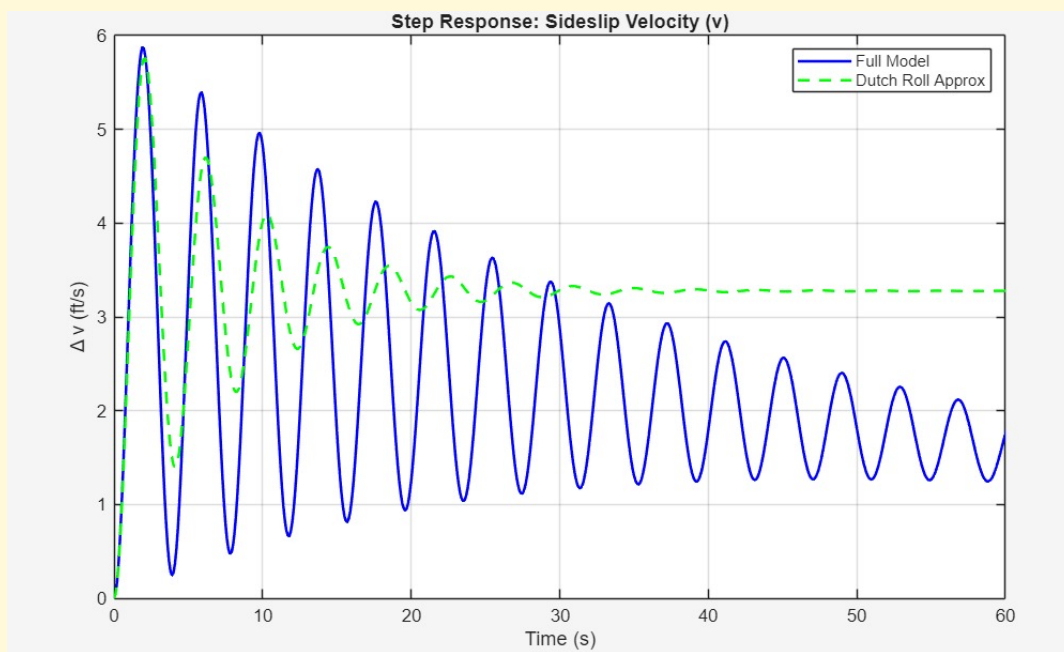


Figure 27: Roll angle (ϕ) step response: full model.

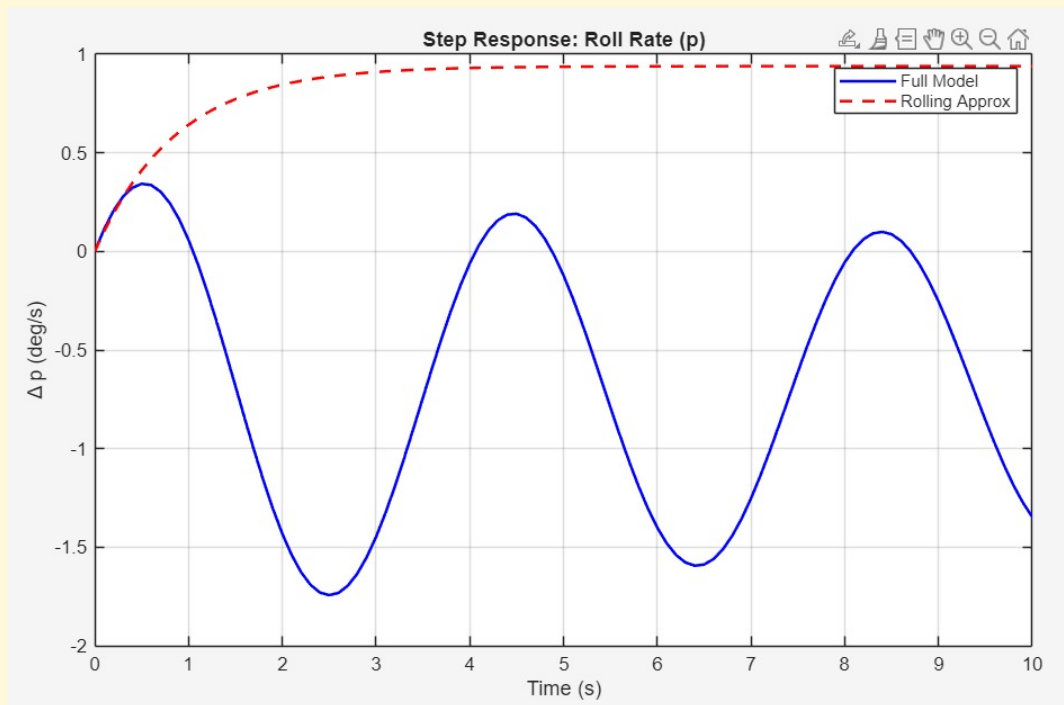


Figure 28: Step response comparison of roll rate change (Δp), comparing the full lateral model

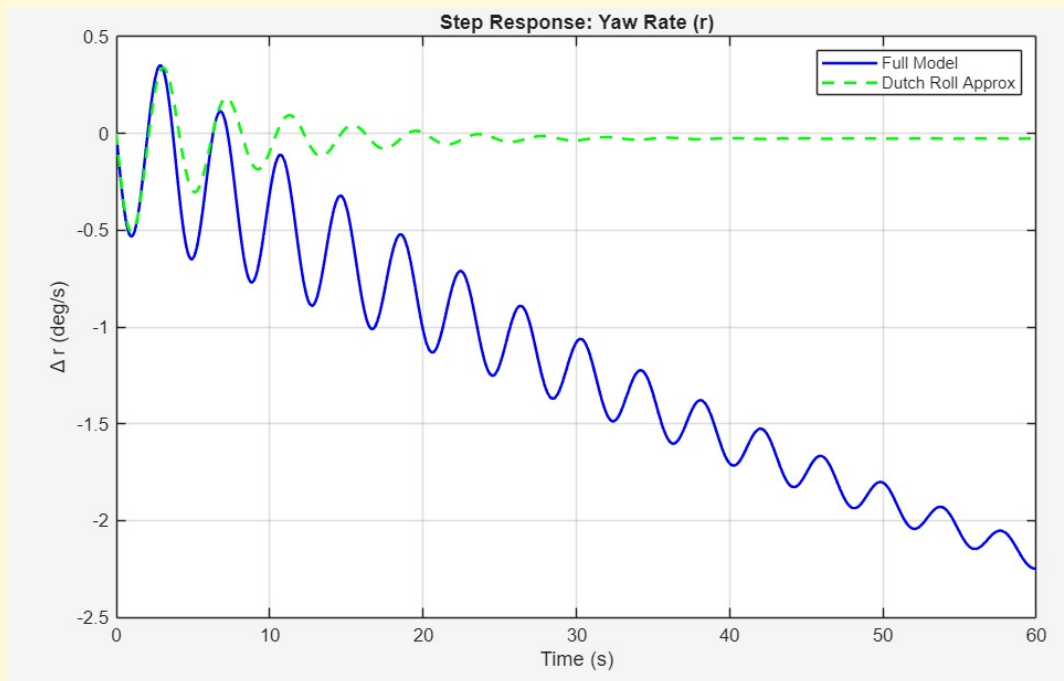


Figure 29: Step response comparison of yaw rate change (Δr) comparing the full lateral model

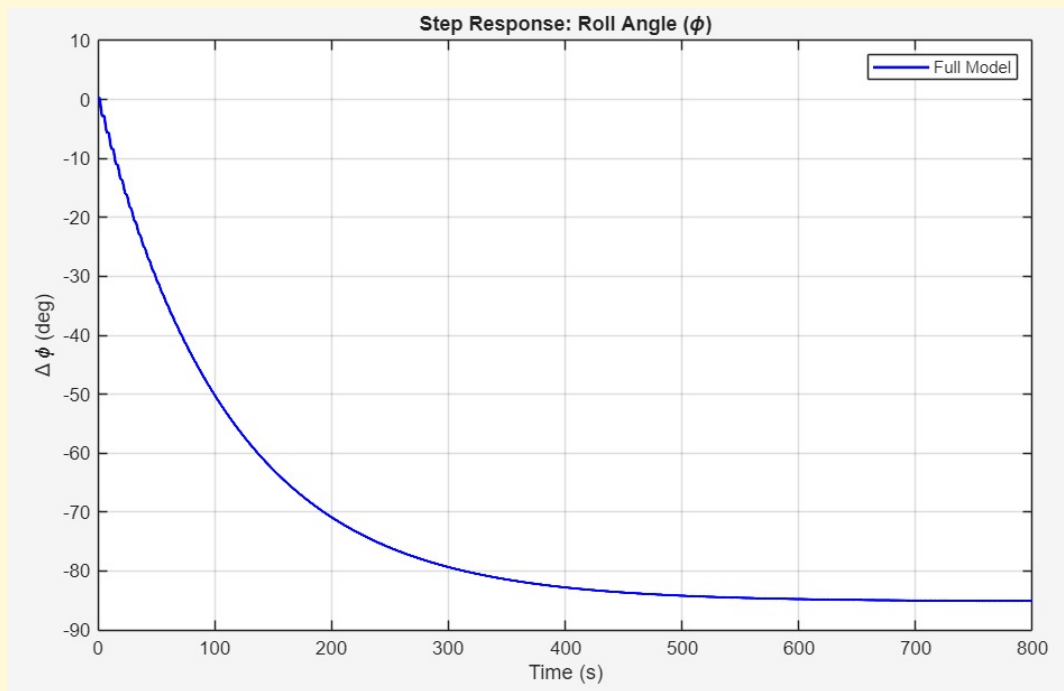


Figure 30: Long-term step response of the roll angle change ($\Delta\phi$) using the full lateral-directional model.

4 Longitudinal and Lateral Autopilot Design (Task 3+)

4.1 Introduction

This section completes Task 3+ by designing longitudinal and lateral attitude autopilots for the aircraft linear models obtained in the previous tasks. The target is to reduce the slow long-period settling behaviour: the phugoid mode in longitudinal motion and the spiral mode in lateral-directional motion.

Autopilot objective. Only single-loop classical attitude-hold autopilots are used. The longitudinal loop commands elevator deflection from pitch-angle error, while the lateral loop commands aileron deflection from roll-angle error.

4.2 Longitudinal Autopilot: Pitch Attitude Hold

4.2.1 State-Space Model

The longitudinal perturbation model uses

$$x_{lon} = \begin{bmatrix} u & w & q & \theta \end{bmatrix}^T, \quad u_{lon} = \delta_e, \quad (34)$$

where u and w are body-axis velocity perturbations, q is pitch rate, θ is pitch angle, and δ_e is elevator deflection.

$$A_{lon} = \begin{bmatrix} 0 & 0.0337 & 0 & -32.174 \\ 0 & -0.4197 & 518.0 & 0 \\ 0 & -0.00158 & -0.420 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B_{lon} = \begin{bmatrix} 2.02 \\ -16.9 \\ -1.037 \\ 0 \end{bmatrix}. \quad (35)$$

The pitch output matrix is

$$C_\theta = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}, \quad D = 0. \quad (36)$$

4.2.2 Transfer Function and Control Loop

The elevator-to-pitch transfer function is obtained from

$$G_\theta(s) = \frac{\theta(s)}{\delta_e(s)} = C_\theta(sI - A_{lon})^{-1}B_{lon}. \quad (37)$$

Using the numerical model,

$$G_\theta(s) = \frac{-1.037s - 0.4085}{s^3 + 0.8397s^2 + 0.9947s}. \quad (38)$$

The feedback loop is

$$e_\theta(s) = \theta_{ref}(s) - \theta(s), \quad \delta_e(s) = C_{\theta c}(s) e_\theta(s), \quad (39)$$

with a PID controller

$$C_{\theta c}(s) = K_p + \frac{K_i}{s} + K_d s. \quad (40)$$

4.2.3 Controller Gains

The tuned longitudinal PID gains are

$$K_p = -0.6500, \quad K_i = -0.0871, \quad K_d = -0.3850. \quad (41)$$

The signs are negative because positive elevator deflection produces a nose-down pitching moment in the adopted convention. Thus, a positive pitch command requires negative elevator feedback action.

The closed-loop pitch command transfer function is

$$T_\theta(s) = \frac{C_{\theta c}(s) G_\theta(s)}{1 + C_{\theta c}(s) G_\theta(s)}. \quad (42)$$

4.2.4 Response Comparison

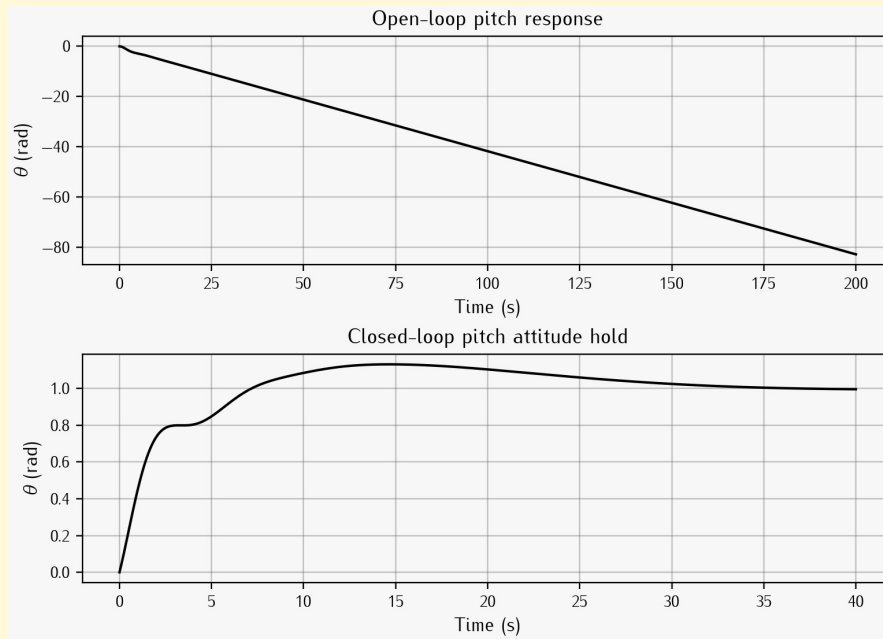


Figure 31: Longitudinal response before and after pitch attitude feedback. The controller damps the phugoid-type long-period response and gives a faster pitch-angle settling behaviour.

4.3 Lateral Autopilot: Roll Attitude Hold

4.3.1 State-Space Model

The lateral-directional perturbation model uses

$$x_{lat} = \begin{bmatrix} v & p & r & \phi \end{bmatrix}^T, \quad u_{lat} = \delta_a, \quad (43)$$

where v is lateral velocity perturbation, p is roll rate, r is yaw rate, ϕ is roll angle, and δ_a is aileron deflection.

$$A_{lat} = \begin{bmatrix} -0.085 & 0 & -518.0 & 32.174 \\ -0.012 & -1.150 & 0.280 & 0 \\ 0.0045 & -0.062 & -0.185 & 0 \\ 0 & 1.0 & 0 & 0 \end{bmatrix}, \quad B_a = \begin{bmatrix} 0 \\ 1.08 \\ -0.03 \\ 0 \end{bmatrix}. \quad (44)$$

The roll output matrix is

$$C_\phi = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}, \quad D = 0. \quad (45)$$

4.3.2 Transfer Function and Control Loop

The aileron-to-roll transfer function is

$$G_\phi(s) = \frac{\phi(s)}{\delta_a(s)} = C_\phi(sI - A_{lat})^{-1}B_a. \quad (46)$$

Using the numerical model,

$$G_\phi(s) = \frac{1.08s^2 + 0.2832s + 2.347}{s^4 + 1.42s^3 + 2.675s^2 + 3.472s + 0.03089}. \quad (47)$$

The roll attitude loop is

$$e_\phi(s) = \phi_{ref}(s) - \phi(s), \quad \delta_a(s) = C_{\phi c}(s) e_\phi(s). \quad (48)$$

A PI controller is sufficient:

$$C_{\phi c}(s) = K_p + \frac{K_i}{s}. \quad (49)$$

4.3.3 Controller Gains

The tuned lateral controller gains are

$$K_p = 0.0191, \quad K_i = 0.000304, \quad K_d = 0. \quad (50)$$

The small denominator constant 0.03089 in $G_\phi(s)$ indicates the slow spiral mode. The controller pulls this long-period behaviour further into the stable left-half plane, reducing the roll settling time.

The closed-loop roll command transfer function is

$$T_\phi(s) = \frac{C_{\phi c}(s) G_\phi(s)}{1 + C_{\phi c}(s) G_\phi(s)}. \quad (51)$$

4.3.4 Response Comparison

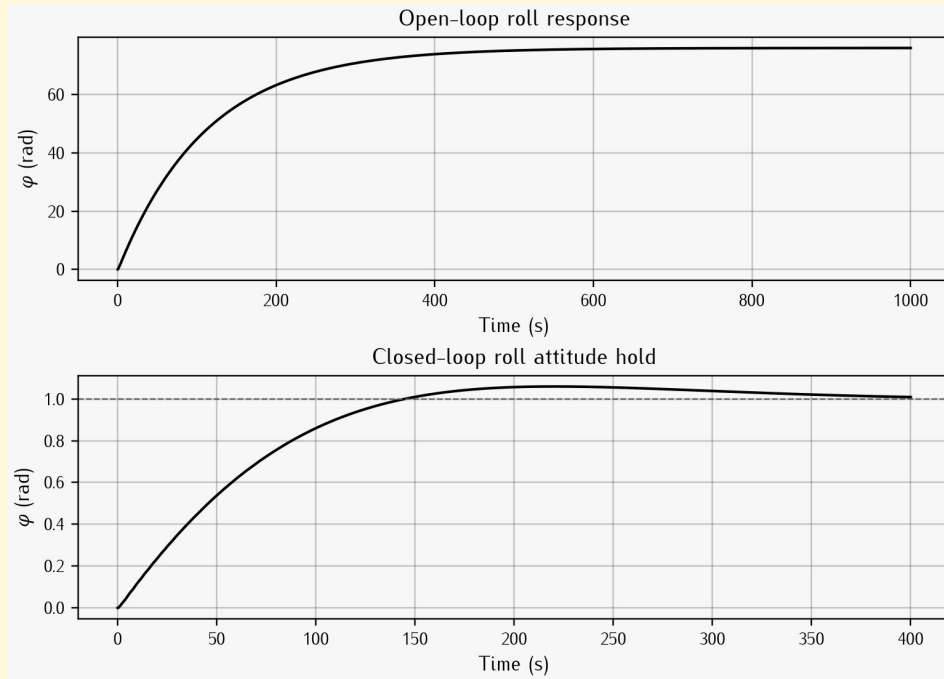


Figure 32: Lateral response before and after roll attitude feedback. The PI autopilot removes the extremely slow spiral-mode settling behaviour and gives a much faster roll- angle response.

4.4 Conclusion

The longitudinal autopilot uses pitch-angle feedback with a PID controller,

$$C_{\theta c}(s) = -0.6500 - \frac{0.0871}{s} - 0.3850s, \quad (52)$$

while the lateral autopilot uses roll-angle feedback with a PI controller,

$$C_{\phi c}(s) = 0.0191 + \frac{0.000304}{s}. \quad (53)$$

Both loops directly target the slow long-period modes. The longitudinal controller damps the phugoid response, and the lateral controller accelerates the spiral-mode roll response.

A MATLAB Code for Full-System Simulation

The following MATLAB code is used to simulate the Boeing 747 longitudinal flight dynamics model discussed in the initial-response and elevator-step sections.

```

1 % Boeing 747 Longitudinal Flight Dynamics Simulation
2 clear; clc; close all;
3
4 % 1. Define System Matrices
5 A = [0,      0.0337,    0,      -32.174;
6      0,      -0.4197,   518,     0;
7      0,      -0.00158, -0.420,    0;
8      0,      0,         1,      0];
9
10 B = [2.02; -16.9; -1.037; 0];
11 C = eye(4);
12 D = zeros(4,1);
13
14 % Create state-space object
15 sys = ss(A, B, C, D);
16 sys.StateName = {'u', 'w', 'q', 'theta'};
17 sys.InputName = {'delta_e'};
18
19 % 2. Define Simulation Parameters
20 u0      = 518;           % Reference steady flight speed (ft/s)
21 t_short = 0:0.01:20;    % Time vector to view Short-Period
22 t_long  = 0:0.1:600;    % Time vector to view Phugoid
23
24 %% --- Initial Condition Response (1 deg Alpha gust) ---
25 x0 = [0; 9.04; 0; 0];
26 [y_initial, t_initial] = initial(sys, x0, t_long);
27
28 u_ini      = y_initial(:, 1);
29 alpha_ini  = y_initial(:, 2) / u0 * (180/pi);
30 q_ini      = y_initial(:, 3) * (180/pi);
31 theta_ini  = y_initial(:, 4) * (180/pi);
32
33 figure('Name', 'Initial Response to 1 deg Alpha Gust');
34 subplot(2,2,1); plot(t_initial, u_ini);      grid on;
35 title('Velocity u (ft/s)');
36 subplot(2,2,2); plot(t_initial, alpha_ini); grid on;
37 title('Angle of Attack alpha (deg)');
38 subplot(2,2,3); plot(t_initial, q_ini);      grid on;
39 title('Pitch Rate q (deg/s)');
40 subplot(2,2,4); plot(t_initial, theta_ini); grid on;
41 title('Pitch Angle theta (deg)');
42
43 %% --- Step Response (1 deg Elevator step) ---
44 delta_e_step = 1 * (pi/180);
45 [y_step, t_step] = step(sys * delta_e_step, t_long);

```

```

46
47 u_step      = y_step(:, 1);
48 alpha_step = y_step(:, 2) / u0 * (180/pi);
49 q_step      = y_step(:, 3) * (180/pi);
50 theta_step = y_step(:, 4) * (180/pi);
51
52 figure('Name', 'Step Response to +1 deg Elevator');
53 subplot(2,2,1); plot(t_step, u_step);      grid on;
54 title('Velocity u (ft/s)');
55 subplot(2,2,2); plot(t_step, alpha_step); grid on;
56 title('Angle of Attack alpha (deg)');
57 subplot(2,2,3); plot(t_step, q_step);      grid on;
58 title('Pitch Rate q (deg/s)');
59 subplot(2,2,4); plot(t_step, theta_step); grid on;
60 title('Pitch Angle theta (deg)');

```

Listing 1: Full-system longitudinal simulation.

B MATLAB Code for Full vs. Reduced Comparison

The following MATLAB code compares the full 4×4 longitudinal model with the short-period and phugoid reduced-order models.

```

1  % Flight Dynamics: Full vs. Reduced Models Comparison
2  clear; clc; close all;
3
4  %% 1. Define Matrices
5
6  % Full System (4x4) - States: [u, w, q, theta]
7  A_full = [-0.008,    0.0337,    0,    -32.174;
8           -0.124,   -0.4197,   518,    0;
9           0,        -0.00158, -0.420,    0;
10          0,         0,         1,     0];
11 B_full = [2.02; -16.9; -1.037; 0];
12 C_full = eye(4); D_full = zeros(4,1);
13 sys_full = ss(A_full, B_full, C_full, D_full);
14
15 % Short-Period Reduced Model (2x2) - States: [w, q]
16 A_sp = [-0.4197,    518;
17         -0.00158, -0.420];
18 B_sp = [-16.9; -1.037];
19 C_sp = eye(2); D_sp = zeros(2,1);
20 sys_sp = ss(A_sp, B_sp, C_sp, D_sp);
21
22 % Phugoid Reduced Model (2x2) - States: [u, theta]
23 A_ph = [-0.008, -32.174;
24         0.00024, 0];
25 B_ph = [2.02; 0];

```

```

26 C_ph = eye(2); D_ph = zeros(2,1);
27 sys_ph = ss(A_ph, B_ph, C_ph, D_ph);
28
29 %% 2. Setup Simulation Parameters
30 t_sim = 0:0.1:400;
31 delta_e = 1 * (pi/180);
32 x0_full = [10; 10; 0; 0.05];
33 x0_sp = [10; 0];
34 x0_ph = [10; 0.05];
35
36 %% 3. Simulate Initial Response
37 [y_f_ini, t] = initial(sys_full, x0_full, t_sim);
38 [y_sp_ini, ~] = initial(sys_sp, x0_sp, t_sim);
39 [y_ph_ini, ~] = initial(sys_ph, x0_ph, t_sim);
40
41 figure('Name', 'Initial Condition Response: Full vs Reduced');
42 subplot(2,2,1); hold on;
43 plot(t, y_f_ini(:,1), 'b-', 'LineWidth', 1.5);
44 plot(t, y_ph_ini(:,1), 'g--', 'LineWidth', 1.5);
45 title('Velocity u (ft/s) - Phugoid Dominant');
46 legend('Full System', 'Phugoid Approx'); grid on;
47
48 subplot(2,2,2); hold on;
49 plot(t, y_f_ini(:,2), 'b-', 'LineWidth', 1.5);
50 plot(t, y_sp_ini(:,1), 'r--', 'LineWidth', 1.5);
51 title('Vertical Vel w (ft/s) - SP Dominant');
52 legend('Full System', 'Short-Period Approx'); grid on; xlim([0 20]);
53
54 subplot(2,2,3); hold on;
55 plot(t, y_f_ini(:,3)*180/pi, 'b-', 'LineWidth', 1.5);
56 plot(t, y_sp_ini(:,2)*180/pi, 'r--', 'LineWidth', 1.5);
57 title('Pitch Rate q (deg/s) - SP Dominant');
58 legend('Full System', 'Short-Period Approx'); grid on; xlim([0 20]);
59
60 subplot(2,2,4); hold on;
61 plot(t, y_f_ini(:,4)*180/pi, 'b-', 'LineWidth', 1.5);
62 plot(t, y_ph_ini(:,2)*180/pi, 'g--', 'LineWidth', 1.5);
63 title('Pitch Angle theta (deg) - Phugoid Dominant');
64 legend('Full System', 'Phugoid Approx'); grid on;
65
66 %% 4. Simulate Step Response
67 [y_f_step, t] = step(sys_full * delta_e, t_sim);
68 [y_sp_step, ~] = step(sys_sp * delta_e, t_sim);
69 [y_ph_step, ~] = step(sys_ph * delta_e, t_sim);
70
71 figure('Name', 'Step Response (1 deg Elev): Full vs Reduced');
72 subplot(2,2,1); hold on;
73 plot(t, y_f_step(:,1), 'b-', 'LineWidth', 1.5);
74 plot(t, y_ph_step(:,1), 'g--', 'LineWidth', 1.5);
75 title('Velocity u (ft/s)');

```

```

76 legend('Full System', 'Phugoid Approx'); grid on;
77
78 subplot(2,2,2); hold on;
79 plot(t, y_f_step(:,2), 'b-', 'LineWidth', 1.5);
80 plot(t, y_sp_step(:,1), 'r--', 'LineWidth', 1.5);
81 title('Vertical Vel w (ft/s)');
82 legend('Full System', 'Short-Period Approx', 'Location', 'best');
83 grid on; xlim([0 30]);
84
85 subplot(2,2,3); hold on;
86 plot(t, y_f_step(:,3)*180/pi, 'b-', 'LineWidth', 1.5);
87 plot(t, y_sp_step(:,2)*180/pi, 'r--', 'LineWidth', 1.5);
88 title('Pitch Rate q (deg/s)');
89 legend('Full System', 'Short-Period Approx'); grid on; xlim([0 30]);
90
91 subplot(2,2,4); hold on;
92 plot(t, y_f_step(:,4)*180/pi, 'b-', 'LineWidth', 1.5);
93 plot(t, y_ph_step(:,2)*180/pi, 'g--', 'LineWidth', 1.5);
94 title('Pitch Angle theta (deg)');
95 legend('Full System', 'Phugoid Approx', 'Location', 'best'); grid on;

```

Listing 2: *Full vs. reduced model comparison.*

C MATLAB Code for Lateral-Directional Dynamics Simulation

```

1 %% Boeing 747 Cruise Lateral Flight Dynamics Automation Script
2 clear; clc; close all;
3
4 %% 1. Matrix Initialization (States: v, p, r, phi | Controls: da, dr)
5 U0 = 518; % Flight path true speed (ft/s)
6 g = 32.174; % Gravity constant (ft/s^2)
7
8 A = [ -0.085, 0.000, -518.000, 32.174;
9       -0.012, -1.150, 0.280, 0.000;
10      0.0045, -0.062, -0.185, 0.000;
11      0.000, 1.000, 0.000, 0.000 ];
12
13 B = [ 0.000, 2.150;
14      1.080, 0.140;
15      -0.030, -0.850;
16      0.000, 0.000 ];
17
18 sys_full = ss(A, B, eye(4), zeros(4,2));
19
20 %% 2. Spectral Analysis
21 [V_full, D_full] = eig(A);

```

```

22 eigs_full = diag(D_full);
23 disp(eigs_full);
24
25 %% 3. Reduced-Order Approximations
26 A_roll = A(2,2); B_roll = B(2,1);
27 sys_roll_approx = ss(A_roll, B_roll, 1, 0);
28
29 A_dr = [A(1,1), A(1,3); A(3,1), A(3,3)];
30 B_dr = [B(1,2); B(3,2)];
31 sys_dr_approx = ss(A_dr, B_dr, eye(2), zeros(2,1));
32
33 num_spiral = A(2,1)*A(3,3) - A(2,3)*A(3,1);
34 den_spiral = A(2,2)*A(3,1) - A(2,1)*A(3,2);
35 A_spiral = (g/U0) * (num_spiral / den_spiral);
36 sys_spiral_approx = ss(A_spiral, 0, 1, 0);
37
38 %% 4. Time-Domain Simulation
39 t_sim = 0:0.05:120;
40 u_null = zeros(length(t_sim), 2);
41 deg2rad = pi/180;
42
43 x0_full = [9.04; 0; 0; 0];
44 [y_gust_full, t] = lsim(sys_full, u_null, t_sim, x0_full);
45 [y_gust_dr, ~] = lsim(sys_dr_approx, zeros(length(t_sim),1), t_sim,
    [9.04; 0]);
46
47 u_step_rudder = [zeros(length(t_sim),1), ones(length(t_sim),1)*deg2rad
    ];
48 u_step_aileron = [ones(length(t_sim),1)*deg2rad, zeros(length(t_sim),1)
    ];
49
50 [y_rudder_full, ~] = lsim(sys_full, u_step_rudder, t_sim);
51 [y_aileron_full, ~] = lsim(sys_full, u_step_aileron, t_sim);
52 [y_rudder_dr, ~] = lsim(sys_dr_approx, ones(length(t_sim),1)*deg2rad
    , t_sim);
53 [y_aileron_roll, ~] = lsim(sys_roll_approx, ones(length(t_sim),1)*
    deg2rad, t_sim);
54
55 %% 5. Visualization
56 figure('Name', 'Gust Response Verification', 'Position', [100, 100,
    800, 600]);
57 subplot(2,1,1);
58 plot(t, y_gust_full(:,1), 'b-', 'LineWidth', 2); hold on;
59 plot(t, y_gust_dr(:,1), 'r--', 'LineWidth', 2);
60 ylabel('Sideslip v (ft/s)'); title('Sideslip under Gust Disturbance');
61 legend('Full Plant', 'Dutch Roll Approx'); grid on;
62
63 subplot(2,1,2);
64 plot(t, y_gust_full(:,3)*180/pi, 'b-', 'LineWidth', 2); hold on;
65 plot(t, y_gust_dr(:,2)*180/pi, 'r--', 'LineWidth', 2);

```

```

66 ylabel('Yaw Rate r (deg/s)'); xlabel('Time (s)');
67 legend('Full Plant', 'Dutch Roll Approx'); grid on;
68
69 figure('Name', 'Aileron Input Verification', 'Position', [200, 200,
    800, 400]);
70 plot(t, y_aileron_full(:,2)*180/pi, 'b-', 'LineWidth', 2); hold on;
71 plot(t, y_aileron_roll(:,1)*180/pi, 'k-', 'LineWidth', 2);
72 ylabel('Roll Rate p (deg/s)'); xlabel('Time (s)');
73 title('Aileron Step Deflection (1 Deg)');
74 legend('Full Plant', 'Rolling Mode Approx'); grid on;

```

Listing 3: Full lateral-directional multi-variable evaluation.

```

1  %% Boeing 747 Lateral-Directional Dynamics - Complete Figure Generator
2  clear; clc; close all;
3
4  results_dir = 'RESULTS';
5  if ~exist(results_dir, 'dir'); mkdir(results_dir); end
6
7  u0 = 518; g = 32.174;
8
9  A_f = [ -0.085,    0.000, -518.000,   32.174;
10         -0.012,   -1.150,    0.280,    0.000;
11         0.0045,  -0.062,   -0.185,    0.000;
12         0.000,    1.000,    0.000,    0.000 ];
13  B_f = [ 0.000,   2.150;
14         1.080,   0.140;
15        -0.030,  -0.850;
16         0.000,   0.000 ];
17
18  sys_full = ss(A_f, B_f, eye(4), zeros(4,2));
19  sys_roll = ss(A_f(2,2), B_f(2,1), 1, 0);
20
21  A_dr = [A_f(1,1), A_f(1,3); A_f(3,1), A_f(3,3)];
22  B_dr = [B_f(1,2); B_f(3,2)];
23  sys_dr = ss(A_dr, B_dr, eye(2), zeros(2,1));
24
25  num_spiral = A_f(2,1)*A_f(3,3) - A_f(2,3)*A_f(3,1);
26  den_spiral = A_f(2,2)*A_f(3,1) - A_f(2,1)*A_f(3,2);
27  A_spiral = (g/u0) * (num_spiral / den_spiral);
28  sys_spiral = ss(A_spiral, 0, 1, 0);
29
30  t_sim = 0:0.1:800;
31  x0_full = [9.04; 0; 0; 5*(pi/180)];
32  da_step = 1*(pi/180); dr_step = 1*(pi/180);
33  u_step = [ones(length(t_sim),1)*da_step, ones(length(t_sim),1)*dr_step
    ];
34
35  [y_f_ini, t] = initial(sys_full, x0_full, t_sim);

```

```

36 [y_roll_ini, ~] = initial(sys_roll, x0_full(2), t_sim);
37 [y_dr_ini, ~] = initial(sys_dr, [x0_full(1); x0_full(3)], t_sim);
38 [y_spiral_ini, ~] = initial(sys_spiral, x0_full(4), t_sim);
39 [y_f_st, ~] = lsim(sys_full, u_step, t_sim);
40 [y_roll_st, ~] = step(sys_roll * da_step, t_sim);
41 [y_dr_st, ~] = step(sys_dr * dr_step, t_sim);
42
43 % Save individual figures
44 f1 = figure; plot(t, y_f_ini(:,1), 'b', t, y_dr_ini(:,1), 'g--', '
    LineWidth', 1.5);
45 grid on; xlim([0 60]); title('Initial Response: Sideslip Velocity (v)')
    ;
46 xlabel('Time (s)'); ylabel('v (ft/s)');
47 legend('Full Model', 'Dutch Roll Approx');
48 saveas(f1, fullfile(results_dir, 'Initial_v.png'));
49
50 f2 = figure; plot(t, rad2deg(y_f_ini(:,2)), 'b', t, rad2deg(y_roll_ini
    (:,1)), 'r--', 'LineWidth', 1.5);
51 grid on; xlim([0 10]); title('Initial Response: Roll Rate (p)');
52 xlabel('Time (s)'); ylabel('p (deg/s)');
53 legend('Full Model', 'Rolling Approx');
54 saveas(f2, fullfile(results_dir, 'Initial_p.png'));
55
56 f3 = figure; plot(t, rad2deg(y_f_ini(:,3)), 'b', t, rad2deg(y_dr_ini
    (:,2)), 'g--', 'LineWidth', 1.5);
57 grid on; xlim([0 60]); title('Initial Response: Yaw Rate (r)');
58 xlabel('Time (s)'); ylabel('r (deg/s)');
59 legend('Full Model', 'Dutch Roll Approx');
60 saveas(f3, fullfile(results_dir, 'Initial_r.png'));
61
62 f4 = figure; plot(t, rad2deg(y_f_ini(:,4)), 'b', t, rad2deg(
    y_spiral_ini(:,1)), 'm:', 'LineWidth', 2);
63 grid on; title('Initial Response: Roll Angle (\phi)');
64 xlabel('Time (s)'); ylabel('\phi (deg)');
65 legend('Full Model', 'Spiral Approx');
66 saveas(f4, fullfile(results_dir, 'Initial_phi.png'));
67
68 f5 = figure; plot(t, y_f_st(:,1), 'b', t, y_dr_st(:,1), 'g--', '
    LineWidth', 1.5);
69 grid on; xlim([0 60]); title('Step Response: Sideslip Velocity (v)');
70 xlabel('Time (s)'); ylabel('\Delta v (ft/s)');
71 legend('Full Model', 'Dutch Roll Approx');
72 saveas(f5, fullfile(results_dir, 'Step_v.png'));
73
74 f6 = figure; plot(t, rad2deg(y_f_st(:,2)), 'b', t, rad2deg(y_roll_st
    (:,1)), 'r--', 'LineWidth', 1.5);
75 grid on; xlim([0 10]); title('Step Response: Roll Rate (p)');
76 xlabel('Time (s)'); ylabel('\Delta p (deg/s)');
77 legend('Full Model', 'Rolling Approx');
78 saveas(f6, fullfile(results_dir, 'Step_p.png'));

```

```

79
80 f7 = figure; plot(t, rad2deg(y_f_st(:,3)), 'b', t, rad2deg(y_dr_st(:,2)
    ), 'g--', 'LineWidth', 1.5);
81 grid on; xlim([0 60]); title('Step Response: Yaw Rate (r)');
82 xlabel('Time (s)'); ylabel('\Delta r (deg/s)');
83 legend('Full Model', 'Dutch Roll Approx');
84 saveas(f7, fullfile(results_dir, 'Step_r.png'));
85
86 f8 = figure; plot(t, rad2deg(y_f_st(:,4)), 'b', 'LineWidth', 1.5);
87 grid on; title('Step Response: Roll Angle (\phi)');
88 xlabel('Time (s)'); ylabel('\Delta \phi (deg)');
89 legend('Full Model');
90 saveas(f8, fullfile(results_dir, 'Step_phi.png'));
91
92 fprintf('Lateral figures saved.\n');

```

Listing 4: *Lateral-directional individual figure generator.*

D MATLAB Code for Autopilot Design and Simulation

The following MATLAB code implements the longitudinal PID and lateral PI autopilots and generates the closed-loop response figures for Section 4.

```

1 %% Task 3+ Autopilot Design: Longitudinal PID + Lateral PI
2 clear; clc; close all;
3
4 results_dir = 'RESULTS';
5 if ~exist(results_dir,'dir'); mkdir(results_dir); end
6
7 %           Longitudinal system
8
9 A_lon = [ 0,      0.0337,    0,      -32.174;
10          0,      -0.4197,  518,      0;
11          0,      -0.00158, -0.420,    0;
12          0,      0,        1,      0 ];
13 B_lon = [2.02; -16.9; -1.037; 0];
14 C_th  = [0 0 0 1];
15
16 sys_lon = ss(A_lon, B_lon, C_th, 0);
17 G_theta = tf(sys_lon);
18
19 % PID controller
20 Kp_lon = -0.6500; Ki_lon = -0.0871; Kd_lon = -0.3850;
21 C_lon = pid(Kp_lon, Ki_lon, Kd_lon);
22
23 % Open- and closed-loop systems
24 OL_lon = C_lon * G_theta;

```

```

24 CL_lon = feedback(OL_lon, 1);
25
26 % Step response comparison
27 t_lon = 0:0.01:60;
28 [y_ol_lon, t_ol] = step(G_theta, t_lon);
29 [y_cl_lon, t_cl] = step(CL_lon, t_lon);
30
31 fig1 = figure('Position', [100 100 800 500]);
32 hold on; grid on;
33 plot(t_ol, y_ol_lon * (180/pi), 'b-', 'LineWidth', 1.5);
34 plot(t_cl, y_cl_lon * (180/pi), 'r--', 'LineWidth', 1.5);
35 xlabel('Time (s)'); ylabel('\theta (deg)');
36 title('Longitudinal Pitch Response: Open-Loop vs Closed-Loop');
37 legend('Open-Loop (no controller)', 'Closed-Loop (PID autopilot)', '
    Location', 'best');
38 saveas(fig1, fullfile(results_dir, 'longitudinal_response.png'));
39
40 % Lateral system
41
42 A_lat = [-0.085, 0, -518.0, 32.174;
43          -0.012, -1.150, 0.280, 0;
44          0.0045, -0.062, -0.185, 0;
45          0, 1.0, 0, 0];
46 B_a = [0; 1.08; -0.03; 0];
47 C_phi = [0 0 0 1];
48
49 sys_lat = ss(A_lat, B_a, C_phi, 0);
50 G_phi = tf(sys_lat);
51
52 % PI controller
53 Kp_lat = 0.0191; Ki_lat = 0.000304;
54 C_lat = pid(Kp_lat, Ki_lat);
55
56 OL_lat = C_lat * G_phi;
57 CL_lat = feedback(OL_lat, 1);
58
59 t_lat = 0:0.1:600;
60 [y_ol_lat, ~] = step(G_phi, t_lat);
61 [y_cl_lat, ~] = step(CL_lat, t_lat);
62
63 fig2 = figure('Position', [150 150 800 500]);
64 hold on; grid on;
65 plot(t_lat, y_ol_lat * (180/pi), 'b-', 'LineWidth', 1.5);
66 plot(t_lat, y_cl_lat * (180/pi), 'r--', 'LineWidth', 1.5);
67 xlabel('Time (s)'); ylabel('\phi (deg)');
68 title('Lateral Roll Response: Open-Loop vs Closed-Loop');
69 legend('Open-Loop (no controller)', 'Closed-Loop (PI autopilot)', '
    Location', 'best');
70 saveas(fig2, fullfile(results_dir, 'lateral_response.png'));

```

```
70  
71 disp('Autopilot figures saved.');
```

Listing 5: *Autopilot design and closed-loop simulation.*

References

- [1] Previous SPC332 Task 1–3 project reports, Boeing 747 flight dynamics data and linear models.
- [2] B. L. Stevens, F. L. Lewis, and E. N. Johnson, *Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems*, 3rd ed., Wiley, 2015.
- [3] R. C. Nelson, *Flight Stability and Automatic Control*, 2nd ed., McGraw-Hill, 1998.